

# KIT SAFETY DATA SHEET



Kit Product Name Monolisa HCV Ag-Ab ULTRA V2, 96 Tests

Kit Catalog Number(s) 72561

Revision date 10-May-2022

## Kit Contents

| Catalog Number(s)   | Product Name                               |
|---------------------|--------------------------------------------|
| 7360G, 5180S        | R8 - Substrat Buffer, 60 mL                |
| 7360J, 5180U, 7361H | R10 - Stopping Solution, 28 ml             |
| 7361A, 7360S, 7360Z | R2 - 20 x Conc. Washing Solution, 70 ml    |
| 7436L, 7436H        | R9 - Chromogen: TMB Solution (11x), (5 mL) |
| 7256B               | R1 - Microplate (12 strips x 8 wells)      |
| 7256C               | R3 - Negative Control (1 ml)               |
| 7256E               | R4 - Ab Positive Control (1,5 ml)          |
| 7256G               | R5a - Ag Positive Control (q.s. ad 1 ml)   |
| 7256H               | R5b - Ag Positive Control Diluent (1 ml)   |
| 7256K               | R6 - Conjugate 1 (15 ml)                   |
| 7256N               | R7 - Conjugate 2 (15 ml)                   |

# SAFETY DATA SHEET



Revision date 03-Mar-2022

Revision Number 1

## 1. Identification

### Product identifier

**Product Name** R8 - Substrat Buffer, 60 mL

### Other means of identification

**Catalog Number(s)** 7360G, 5180S

### Recommended use of the chemical and restrictions on use

**Recommended use** In vitro diagnostic Restricted to professional users

### Details of the supplier of the safety data sheet

#### Corporate Headquarters

Bio-Rad Laboratories Inc.  
1000 Alfred Nobel Drive  
Hercules, CA 94547  
USA

#### Manufacturer Address

Bio-Rad  
3 boulevard Raymond Poincaré  
92430 Marnes-la-Coquette  
France  
e-mail: fds-msds.fr@bio-rad.com

#### Legal Entity / Contact Address

Bio-Rad Laboratories  
Clinical Diagnostics  
4000 Alfred Nobel Drive  
Hercules, California 94547  
USA

**Technical Service** 1-800-224-6723  
TechSupport.USSD@bio-rad.com

### Emergency telephone number

**24 Hour Emergency Phone Number** CHEMTREC USA: 1 (800) 424-9300

## 2. Hazard(s) identification

### Classification

This chemical is not considered hazardous by the 2012 OSHA Hazard Communication Standard (29 CFR 1910.1200)

### Hazards not otherwise classified (HNOC)

Not applicable

### Label elements

#### Hazard statements

This chemical is not considered hazardous by the 2012 OSHA Hazard Communication Standard (29 CFR 1910.1200)

**Appearance** aqueous solution

**Physical state** Liquid

**Odor** Odorless

### Other information

No information available.

### 3. Composition/information on ingredients

**Substance**

Not applicable.

**Mixture**

| Chemical name  | CAS No   | Weight-% | Trade secret |
|----------------|----------|----------|--------------|
| Citric acid    | 77-92-9  | 1 - 2.5  | *            |
| Sodium acetate | 127-09-3 | 1 - 2.5  | *            |

\*The exact percentage (concentration) of composition has been withheld as a trade secret.

### 4. First-aid measures

**Description of first aid measures**

|              |                                                                                                                      |
|--------------|----------------------------------------------------------------------------------------------------------------------|
| Inhalation   | Remove to fresh air.                                                                                                 |
| Eye contact  | Rinse thoroughly with plenty of water for at least 15 minutes, lifting lower and upper eyelids. Consult a physician. |
| Skin contact | Wash skin with soap and water.                                                                                       |
| Ingestion    | Rinse mouth.                                                                                                         |

**Most important symptoms and effects, both acute and delayed**

Symptoms No information available.

**Indication of any immediate medical attention and special treatment needed**

Note to physicians Treat symptomatically.

### 5. Fire-fighting measures

**Suitable Extinguishing Media** Use extinguishing measures that are appropriate to local circumstances and the surrounding environment.

**Specific hazards arising from the chemical** No information available.

**Explosion data**

**Sensitivity to mechanical impact** None.

**Sensitivity to static discharge** None.

**Special protective equipment and precautions for fire-fighters** Firefighters should wear self-contained breathing apparatus and full firefighting turnout gear. Use personal protection equipment.

### 6. Accidental release measures

**Personal precautions, protective equipment and emergency procedures**

**Personal precautions** Ensure adequate ventilation.

**Methods and material for containment and cleaning up**

**Methods for containment** Prevent further leakage or spillage if safe to do so.

**Methods for cleaning up** Pick up and transfer to properly labeled containers.

**7. Handling and storage****Precautions for safe handling**

**Advice on safe handling** Handle in accordance with good industrial hygiene and safety practice.

**Conditions for safe storage, including any incompatibilities**

**Storage Conditions** Keep container tightly closed in a dry and well-ventilated place.

**8. Exposure controls/personal protection****Control parameters**

**Exposure Limits** The following ingredients are the only ingredients of the product above the cut-off level (or level that contributes to the hazard classification of the mixture) which have an exposure limit applicable in the region for which this safety data sheet is intended or other recommended limit. At this time, the other relevant constituents have no known exposure limits from the sources listed here.

**Appropriate engineering controls**

**Engineering controls** Showers  
Eyewash stations  
Ventilation systems.

**Individual protection measures, such as personal protective equipment**

**Eye/face protection** No special protective equipment required.

**Skin and body protection** No special protective equipment required.

**Respiratory protection** No protective equipment is needed under normal use conditions. If exposure limits are exceeded or irritation is experienced, ventilation and evacuation may be required.

**General hygiene considerations** Handle in accordance with good industrial hygiene and safety practice.

**9. Physical and chemical properties****Information on basic physical and chemical properties**

**Physical state** Liquid  
**Appearance** aqueous solution  
**Color** colorless  
**Odor** Odorless  
**Odor threshold** No information available

| <u>Property</u>                        | <u>Values</u>            | <u>Remarks • Method</u> |
|----------------------------------------|--------------------------|-------------------------|
| pH                                     |                          |                         |
| Melting point / freezing point         | No data available        | None known              |
| Boiling point / boiling range          | No data available        | None known              |
| Flash point                            | No data available        | None known              |
| Evaporation rate                       | No data available        | None known              |
| Flammability (solid, gas)              | No data available        | None known              |
| Flammability Limit in Air              |                          | None known              |
| Upper flammability or explosive limits | No data available        |                         |
| Lower flammability or explosive limits | No data available        |                         |
| Vapor pressure                         | No data available        | None known              |
| Vapor density                          | No data available        | None known              |
| Relative density                       | No data available        | None known              |
| Water solubility                       | Miscible in water        |                         |
| Solubility(ies)                        | No data available        | None known              |
| Partition coefficient                  | No data available        | None known              |
| Autoignition temperature               | 1010 °C / 1850 °F        |                         |
| Decomposition temperature              |                          | None known              |
| Kinematic viscosity                    | No data available        | None known              |
| Dynamic viscosity                      | No data available        | None known              |
| <u>Other information</u>               |                          |                         |
| Explosive properties                   | No information available |                         |
| Oxidizing properties                   | No information available |                         |
| Softening point                        | No information available |                         |
| Molecular weight                       | No information available |                         |
| VOC Content (%)                        | No information available |                         |
| Liquid Density                         | No information available |                         |
| Bulk density                           | No information available |                         |

## 10. Stability and reactivity

|                                    |                                           |
|------------------------------------|-------------------------------------------|
| Reactivity                         | No information available.                 |
| Chemical stability                 | Stable under normal conditions.           |
| Possibility of hazardous reactions | None under normal processing.             |
| Conditions to avoid                | None known based on information supplied. |
| Incompatible materials             | None known based on information supplied. |
| Hazardous decomposition products   | None known based on information supplied. |

## 11. Toxicological information

### Information on likely routes of exposure

|              |                                                                   |
|--------------|-------------------------------------------------------------------|
| Inhalation   | Specific test data for the substance or mixture is not available. |
| Eye contact  | Specific test data for the substance or mixture is not available. |
| Skin contact | Specific test data for the substance or mixture is not available. |
| Ingestion    | Specific test data for the substance or mixture is not available. |

### Symptoms related to the physical, chemical and toxicological characteristics

|          |                           |
|----------|---------------------------|
| Symptoms | No information available. |
|----------|---------------------------|

**Acute toxicity****Numerical measures of toxicity**

The following values are calculated based on chapter 3.1 of the GHS document

|                               |                   |
|-------------------------------|-------------------|
| ATEmix (oral)                 | 82,278.5183 mg/kg |
| ATEmix (dermal)               | 96,611.40 mg/kg   |
| ATEmix (inhalation-dust/mist) | 552.00 mg/l       |

**Component Information**

| Chemical name              | Oral LD50            | Dermal LD50          | Inhalation LC50                   |
|----------------------------|----------------------|----------------------|-----------------------------------|
| Citric acid<br>77-92-9     | = 3 g/kg ( Rat )     | > 2000 mg/kg ( Rat ) | -                                 |
| Sodium acetate<br>127-09-3 | = 3530 mg/kg ( Rat ) | > 10 g/kg ( Rabbit ) | > 30 g/m <sup>3</sup> ( Rat ) 1 h |

**Delayed and immediate effects as well as chronic effects from short and long-term exposure**

**Skin corrosion/irritation** No information available.

**Serious eye damage/eye irritation** No information available.

**Respiratory or skin sensitization** No information available.

**Germ cell mutagenicity** No information available.

**Carcinogenicity** No information available.

**Reproductive toxicity** No information available.

**STOT - single exposure** No information available.

**STOT - repeated exposure** No information available.

**Aspiration hazard** No information available.

**Other adverse effects** No information available.

**Interactive effects** No information available.

## 12. Ecological information

**Ecotoxicity**

| Chemical name              | Algae/aquatic plants | Fish                                               | Toxicity to microorganisms | Crustacea                                    |
|----------------------------|----------------------|----------------------------------------------------|----------------------------|----------------------------------------------|
| Citric acid<br>77-92-9     | -                    | LC50: =1516mg/L (96h, <i>Lepomis macrochirus</i> ) | -                          | -                                            |
| Sodium acetate<br>127-09-3 | -                    | LC50: >100mg/L (96h, <i>Danio rerio</i> )          | -                          | EC50: >1000mg/L (48h, <i>Daphnia magna</i> ) |

**Persistence and degradability** No information available.

**Bioaccumulation****Component Information**

| Chemical name          | Partition coefficient |
|------------------------|-----------------------|
| Citric acid<br>77-92-9 | -1.72                 |

**Other adverse effects** No information available.

**13. Disposal considerations****Waste treatment methods**

**Waste from residues/unused products** Dispose of in accordance with local regulations. Dispose of waste in accordance with environmental legislation.

**Contaminated packaging** Do not reuse empty containers.

**14. Transport information**

**DOT** Not regulated

**TDG** Not regulated

**MEX** Not regulated

**IATA** Not regulated

**IMDG** Not regulated

**15. Regulatory information**

**International Inventories** Contact supplier for inventory compliance status

**US Federal Regulations****SARA 313**

Section 313 of Title III of the Superfund Amendments and Reauthorization Act of 1986 (SARA). This product does not contain any chemicals which are subject to the reporting requirements of the Act and Title 40 of the Code of Federal Regulations, Part 372.

**SARA 311/312 Hazard Categories**

Should this product meet EPCRA 311/312 Tier reporting criteria at 40 CFR 370, refer to Section 2 of this SDS for appropriate classifications.

**CWA (Clean Water Act)**

This product contains the following substances which are regulated pollutants pursuant to the Clean Water Act (40 CFR 122.21 and 40 CFR 122.42).

**CERCLA**

This material, as supplied, does not contain any substances regulated as hazardous substances under the Comprehensive Environmental Response Compensation and Liability Act (CERCLA) (40 CFR 302) or the Superfund Amendments and Reauthorization Act (SARA) (40 CFR 355). There may be specific reporting requirements at the local, regional, or state level

pertaining to releases of this material.

### US State Regulations

#### California Proposition 65

This product does not contain any Proposition 65 chemicals.

### U.S. State Right-to-Know Regulations

| Chemical name                              | New Jersey | Massachusetts | Pennsylvania |
|--------------------------------------------|------------|---------------|--------------|
| Water<br>7732-18-5                         | -          | -             | X            |
| Dimethyl sulfoxide<br>67-68-5              | X          | -             | -            |
| Hydrogen peroxide<br>7722-84-1             | X          | X             | X            |
| Ethylenediaminetetraacetic acid<br>60-00-4 | X          | X             | X            |

### U.S. EPA Label Information

**EPA Pesticide Registration Number** Not applicable

## 16. Other information

|             |                         |                       |                           |                              |
|-------------|-------------------------|-----------------------|---------------------------|------------------------------|
| <b>NFPA</b> | <b>Health hazards</b> 0 | <b>Flammability</b> 0 | <b>Instability</b> 0      | <b>Special hazards</b> -     |
| <b>HMIS</b> | <b>Health hazards</b> 0 | <b>Flammability</b> 0 | <b>Physical hazards</b> 0 | <b>Personal protection</b> X |

### Key or legend to abbreviations and acronyms used in the safety data sheet

#### **Legend Section 8: EXPOSURE CONTROLS/PERSONAL PROTECTION**

|         |                             |      |                                  |
|---------|-----------------------------|------|----------------------------------|
| TWA     | TWA (time-weighted average) | STEL | STEL (Short Term Exposure Limit) |
| Ceiling | Maximum limit value         | *    | Skin designation                 |

### **Key literature references and sources for data used to compile the SDS**

Agency for Toxic Substances and Disease Registry (ATSDR)  
 U.S. Environmental Protection Agency ChemView Database  
 European Food Safety Authority (EFSA)  
 EPA (Environmental Protection Agency)  
 Acute Exposure Guideline Level(s) (AEGL(s))  
 U.S. Environmental Protection Agency Federal Insecticide, Fungicide, and Rodenticide Act  
 U.S. Environmental Protection Agency High Production Volume Chemicals  
 Food Research Journal  
 Hazardous Substance Database  
 International Uniform Chemical Information Database (IUCLID)  
 National Institute of Technology and Evaluation (NITE)  
 Australia National Industrial Chemicals Notification and Assessment Scheme (NICNAS)  
 NIOSH (National Institute for Occupational Safety and Health)  
 National Library of Medicine's ChemID Plus (NLM CIP)  
 National Library of Medicine's PubMed database (NLM PUBMED)  
 National Toxicology Program (NTP)  
 New Zealand's Chemical Classification and Information Database (CCID)  
 Organization for Economic Co-operation and Development Environment, Health, and Safety Publications  
 Organization for Economic Co-operation and Development High Production Volume Chemicals Program  
 Organization for Economic Co-operation and Development Screening Information Data Set  
 World Health Organization

**Revision date** 03-Mar-2022

**Revision Note** Significant changes throughout SDS. Review all sections.

### Disclaimer

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the



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date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text.

**End of Safety Data Sheet**

# SAFETY DATA SHEET



Revision date 02-Mar-2022

Revision Number 1

## 1. Identification

### Product identifier

**Product Name** R10 - Stopping Solution, 28 ml

### Other means of identification

**Catalog Number(s)** 7360J, 5180U, 7361H  
**UN/ID no** UN2796

### Recommended use of the chemical and restrictions on use

**Recommended use** In vitro diagnostic Restricted to professional users

### Details of the supplier of the safety data sheet

#### Corporate Headquarters

Bio-Rad Laboratories Inc.  
1000 Alfred Nobel Drive  
Hercules, CA 94547  
USA

#### Manufacturer Address

Bio-Rad  
3 boulevard Raymond Poincaré  
92430 Marnes-la-Coquette  
France  
e-mail: fds-msds.fr@bio-rad.com

#### Legal Entity / Contact Address

Bio-Rad Laboratories  
Clinical Diagnostics  
4000 Alfred Nobel Drive  
Hercules, California 94547  
USA

**Technical Service** 1-800-224-6723  
TechSupport.USSD@bio-rad.com

### Emergency telephone number

**24 Hour Emergency Phone Number** CHEMTREC USA: 1 (800) 424-9300

## 2. Hazard(s) identification

### Classification

|                                   |            |
|-----------------------------------|------------|
| Skin corrosion/irritation         | Category 1 |
| Serious eye damage/eye irritation | Category 1 |

### Hazards not otherwise classified (HNOC)

Not applicable

### Label elements

**Danger**

#### Hazard statements

Causes severe skin burns and eye damage



|                                    |                              |                 |
|------------------------------------|------------------------------|-----------------|
| <b>Appearance</b> aqueous solution | <b>Physical state</b> Liquid | <b>Odor</b> Low |
|------------------------------------|------------------------------|-----------------|

**Precautionary Statements - Prevention**

Do not breathe dusts or mists  
 Wash face, hands and any exposed skin thoroughly after handling  
 Wear protective gloves/protective clothing/eye protection/face protection

**Precautionary Statements - Response**

Immediately call a POISON CENTER or doctor  
 IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing  
 Immediately call a POISON CENTER or doctor  
 IF ON SKIN (or hair): Take off immediately all contaminated clothing. Rinse skin with water/ shower  
 Wash contaminated clothing before reuse  
 IF INHALED: Remove person to fresh air and keep comfortable for breathing  
 Immediately call a POISON CENTER or doctor  
 IF SWALLOWED: Rinse mouth. Do NOT induce vomiting

**Precautionary Statements - Disposal**

Dispose of contents/container in accordance with local, regional, national, and international regulations as applicable

**Other information**

No information available.

### 3. Composition/information on ingredients

**Substance**

Not applicable.

**Mixture**

| Chemical name | CAS No    | Weight-% | Trade secret |
|---------------|-----------|----------|--------------|
| Sulfuric acid | 7664-93-9 | 2.5 - 5  | *            |

\*The exact percentage (concentration) of composition has been withheld as a trade secret.

### 4. First-aid measures

**Description of first aid measures**

|                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|-------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>General advice</b>                     | Immediate medical attention is required. Show this safety data sheet to the doctor in attendance.                                                                                                                                                                                                                                                                                                                                                                              |
| <b>Inhalation</b>                         | Remove to fresh air. If breathing has stopped, give artificial respiration. Get medical attention immediately. Do not use mouth-to-mouth method if victim ingested or inhaled the substance; give artificial respiration with the aid of a pocket mask equipped with a one-way valve or other proper respiratory medical device. If breathing is difficult, (trained personnel should) give oxygen. Delayed pulmonary edema may occur. Get immediate medical advice/attention. |
| <b>Eye contact</b>                        | Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Keep eye wide open while rinsing. Do not rub affected area. Remove contact lenses, if present and easy to do. Continue rinsing. Get immediate medical advice/attention.                                                                                                                                                                                                               |
| <b>Skin contact</b>                       | Wash off immediately with soap and plenty of water while removing all contaminated clothes and shoes. Get immediate medical advice/attention.                                                                                                                                                                                                                                                                                                                                  |
| <b>Ingestion</b>                          | Do NOT induce vomiting. Rinse mouth. Never give anything by mouth to an unconscious person. Get immediate medical advice/attention.                                                                                                                                                                                                                                                                                                                                            |
| <b>Self-protection of the first aider</b> | Ensure that medical personnel are aware of the material(s) involved, take precautions to                                                                                                                                                                                                                                                                                                                                                                                       |

protect themselves and prevent spread of contamination. Avoid contact with skin, eyes or clothing. Avoid direct contact with skin. Use barrier to give mouth-to-mouth resuscitation. Wear personal protective clothing (see section 8).

**Most important symptoms and effects, both acute and delayed**

**Symptoms** Burning sensation.

**Indication of any immediate medical attention and special treatment needed**

**Note to physicians** Product is a corrosive material. Use of gastric lavage or emesis is contraindicated. Possible perforation of stomach or esophagus should be investigated. Do not give chemical antidotes. Asphyxia from glottal edema may occur. Marked decrease in blood pressure may occur with moist rales, frothy sputum, and high pulse pressure.

## 5. Fire-fighting measures

**Suitable Extinguishing Media** Use extinguishing measures that are appropriate to local circumstances and the surrounding environment.

**Specific hazards arising from the chemical** The product causes burns of eyes, skin and mucous membranes. Thermal decomposition can lead to release of irritating gases and vapors.

**Explosion data**  
**Sensitivity to mechanical impact** None.

**Sensitivity to static discharge** None.

**Special protective equipment and precautions for fire-fighters** Firefighters should wear self-contained breathing apparatus and full firefighting turnout gear. Use personal protection equipment.

## 6. Accidental release measures

**Personal precautions, protective equipment and emergency procedures**

**Personal precautions** Attention! Corrosive material. Avoid contact with skin, eyes or clothing. Ensure adequate ventilation. Use personal protective equipment as required. Evacuate personnel to safe areas. Keep people away from and upwind of spill/leak.

**Other information** Refer to protective measures listed in Sections 7 and 8.

**Methods and material for containment and cleaning up**

**Methods for containment** Prevent further leakage or spillage if safe to do so.

**Methods for cleaning up** Pick up and transfer to properly labeled containers.

## 7. Handling and storage

**Precautions for safe handling**

**Advice on safe handling** Handle in accordance with good industrial hygiene and safety practice. Avoid contact with skin, eyes or clothing. In case of insufficient ventilation, wear suitable respiratory equipment. Handle product only in closed system or provide appropriate exhaust ventilation. Do not eat, drink or smoke when using this product. Take off contaminated clothing and wash before reuse.

**Conditions for safe storage, including any incompatibilities**

**Storage Conditions**

Keep containers tightly closed in a dry, cool and well-ventilated place. Protect from moisture. Store locked up. Keep out of the reach of children. Store away from other materials. Store according to product and label instructions.

## 8. Exposure controls/personal protection

**Control parameters****Exposure Limits**

| Chemical name              | ACGIH TLV                                                 | OSHA PEL                                                       | NIOSH                                                  |
|----------------------------|-----------------------------------------------------------|----------------------------------------------------------------|--------------------------------------------------------|
| Sulfuric acid<br>7664-93-9 | TWA: 0.2 mg/m <sup>3</sup> thoracic<br>particulate matter | TWA: 1 mg/m <sup>3</sup><br>(vacated) TWA: 1 mg/m <sup>3</sup> | IDLH: 15 mg/m <sup>3</sup><br>TWA: 1 mg/m <sup>3</sup> |

**Appropriate engineering controls****Engineering controls**

Showers  
Eyewash stations  
Ventilation systems.

**Individual protection measures, such as personal protective equipment****Eye/face protection**

Tight sealing safety goggles. Face protection shield.

**Hand protection**

Wear suitable gloves. Impervious gloves.

**Skin and body protection**

Wear suitable protective clothing. Long sleeved clothing. Chemical resistant apron.

**Respiratory protection**

No protective equipment is needed under normal use conditions. If exposure limits are exceeded or irritation is experienced, ventilation and evacuation may be required.

**General hygiene considerations**

Avoid contact with skin, eyes or clothing. Wear suitable gloves and eye/face protection. Do not eat, drink or smoke when using this product. Remove and wash contaminated clothing and gloves, including the inside, before re-use. Contaminated work clothing should not be allowed out of the workplace. Regular cleaning of equipment, work area and clothing is recommended. Wash hands before breaks and immediately after handling the product.

## 9. Physical and chemical properties

**Information on basic physical and chemical properties**

|                |                          |
|----------------|--------------------------|
| Physical state | Liquid                   |
| Appearance     | aqueous solution         |
| Color          | colorless                |
| Odor           | Low                      |
| Odor threshold | No information available |

**Property****Values****Remarks • Method**

|                                        |                   |            |
|----------------------------------------|-------------------|------------|
| pH                                     | < 2               |            |
| Melting point / freezing point         | No data available | None known |
| Boiling point / boiling range          | No data available | None known |
| Flash point                            | No data available | None known |
| Evaporation rate                       | No data available | None known |
| Flammability (solid, gas)              | No data available | None known |
| Flammability Limit in Air              |                   | None known |
| Upper flammability or explosive limits | No data available |            |
| Lower flammability or explosive limits | No data available |            |

|                           |                   |            |
|---------------------------|-------------------|------------|
| Vapor pressure            | No data available | None known |
| Vapor density             | No data available | None known |
| Relative density          | No data available | None known |
| Water solubility          | Miscible in water |            |
| Solubility(ies)           | No data available | None known |
| Partition coefficient     | No data available | None known |
| Autoignition temperature  | No data available | None known |
| Decomposition temperature |                   | None known |
| Kinematic viscosity       | No data available | None known |
| Dynamic viscosity         | No data available | None known |

**Other information**

|                      |                          |
|----------------------|--------------------------|
| Explosive properties | No information available |
| Oxidizing properties | No information available |
| Softening point      | No information available |
| Molecular weight     | No information available |
| VOC Content (%)      | No information available |
| Liquid Density       | No information available |
| Bulk density         | No information available |

## 10. Stability and reactivity

|                                    |                                                     |
|------------------------------------|-----------------------------------------------------|
| Reactivity                         | No information available.                           |
| Chemical stability                 | Stable under normal conditions.                     |
| Possibility of hazardous reactions | None under normal processing.                       |
| Conditions to avoid                | Exposure to air or moisture over prolonged periods. |
| Incompatible materials             | Acids. Bases. Oxidizing agent.                      |
| Hazardous decomposition products   | None known based on information supplied.           |

## 11. Toxicological information

**Information on likely routes of exposure****Product Information**

|                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|---------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Inhalation</b>   | Specific test data for the substance or mixture is not available. Corrosive by inhalation. (based on components). Inhalation of corrosive fumes/gases may cause coughing, choking, headache, dizziness, and weakness for several hours. Pulmonary edema may occur with tightness in the chest, shortness of breath, bluish skin, decreased blood pressure, and increased heart rate. Inhaled corrosive substances can lead to a toxic edema of the lungs. Pulmonary edema can be fatal.                                   |
| <b>Eye contact</b>  | Specific test data for the substance or mixture is not available. Causes serious eye damage. (based on components). Corrosive to the eyes and may cause severe damage including blindness. May cause irreversible damage to eyes.                                                                                                                                                                                                                                                                                         |
| <b>Skin contact</b> | Specific test data for the substance or mixture is not available. Corrosive. (based on components). Causes burns.                                                                                                                                                                                                                                                                                                                                                                                                         |
| <b>Ingestion</b>    | Specific test data for the substance or mixture is not available. Causes burns. (based on components). Ingestion causes burns of the upper digestive and respiratory tracts. May cause severe burning pain in the mouth and stomach with vomiting and diarrhea of dark blood. Blood pressure may decrease. Brownish or yellowish stains may be seen around the mouth. Swelling of the throat may cause shortness of breath and choking. May cause lung damage if swallowed. May be fatal if swallowed and enters airways. |

**Symptoms related to the physical, chemical and toxicological characteristics**

**Symptoms** Redness. Burning. May cause blindness. Coughing and/ or wheezing.

**Acute toxicity**

**Numerical measures of toxicity**

**Component Information**

| Chemical name              | Oral LD50            | Dermal LD50 | Inhalation LC50          |
|----------------------------|----------------------|-------------|--------------------------|
| Sulfuric acid<br>7664-93-9 | = 2140 mg/kg ( Rat ) | -           | = 0.375 mg/L ( Rat ) 4 h |

**Delayed and immediate effects as well as chronic effects from short and long-term exposure**

**Skin corrosion/irritation** Classification based on data available for ingredients. Causes burns.

**Serious eye damage/eye irritation** Classification based on data available for ingredients. Risk of serious damage to eyes. Causes burns.

**Respiratory or skin sensitization** No information available.

**Germ cell mutagenicity** No information available.

**Carcinogenicity** No information available.

The table below indicates whether each agency has listed any ingredient as a carcinogen.

| Chemical name              | ACGIH | IARC    | NTP   | OSHA |
|----------------------------|-------|---------|-------|------|
| Sulfuric acid<br>7664-93-9 | A2    | Group 1 | Known | X    |

**Legend**

**ACGIH (American Conference of Governmental Industrial Hygienists)**

A2 - Suspected Human Carcinogen

**IARC (International Agency for Research on Cancer)**

Group 1 - Carcinogenic to Humans

**NTP (National Toxicology Program)**

Known - Known Carcinogen

**OSHA (Occupational Safety and Health Administration of the US Department of Labor)**

X - Present

**Reproductive toxicity** No information available.

**STOT - single exposure** No information available.

**STOT - repeated exposure** No information available.

**Target organ effects** Respiratory system, Eyes, Skin, Teeth.

**Aspiration hazard** No information available.

**Other adverse effects** No information available.

**Interactive effects** No information available.

## 12. Ecological information

### Ecotoxicity

| Chemical name              | Algae/aquatic plants | Fish                                    | Toxicity to microorganisms | Crustacea |
|----------------------------|----------------------|-----------------------------------------|----------------------------|-----------|
| Sulfuric acid<br>7664-93-9 | -                    | LC50: >500mg/L (96h, Brachydanio rerio) | -                          | -         |

**Persistence and degradability** No information available.

**Bioaccumulation** There is no data for this product.

**Other adverse effects** No information available.

## 13. Disposal considerations

### Waste treatment methods

**Waste from residues/unused products** Dispose of in accordance with local regulations. Dispose of waste in accordance with environmental legislation.

**Contaminated packaging** Do not reuse empty containers.

**US EPA Waste Number** D002.

**California Hazardous Waste Status** This product contains one or more substances that are listed with the State of California as a hazardous waste.

## 14. Transport information

### DOT

|                                     |                                                                  |
|-------------------------------------|------------------------------------------------------------------|
| UN/ID no                            | UN2796                                                           |
| Proper shipping name                | SULFURIC ACID SOLUTION                                           |
| Transport hazard class(es)          | 8                                                                |
| Packing group                       | II                                                               |
| Reportable Quantity (RQ)            | (Sulfuric acid: RQ (kg)= 454.00) Sulfuric acid: RQ (lb)= 1000.00 |
| Reportable quantity kg (calculated) | Sulfuric acid: RQ (kg)= 9458.00                                  |
| Special Provisions                  | 386, A3, A7, B2, B15, IB2, N6, N34, T8, TP2                      |
| DOT Marine Pollutant                | NP                                                               |
| Description                         | UN2796, SULFURIC ACID SOLUTION, 8, II                            |
| Emergency Response Guide Number     | 157                                                              |

### MEX

|                            |                                        |
|----------------------------|----------------------------------------|
| UN/ID no                   | UN2796                                 |
| Proper shipping name       | SULPHURIC ACID SOLUTION                |
| Transport hazard class(es) | 8                                      |
| Packing group              | II                                     |
| Description                | UN2796, SULPHURIC ACID SOLUTION, 8, II |

### IATA

|                            |                         |
|----------------------------|-------------------------|
| UN number or ID number     | UN2796                  |
| UN proper shipping name    | Sulphuric acid solution |
| Transport hazard class(es) | 8                       |



|                      |                                        |
|----------------------|----------------------------------------|
| <b>Packing group</b> | II                                     |
| <b>Description</b>   | UN2796, Sulphuric acid solution, 8, II |

**IMDG**

|                                   |                                        |
|-----------------------------------|----------------------------------------|
| <b>UN number or ID number</b>     | UN2796                                 |
| <b>UN proper shipping name</b>    | SULPHURIC ACID SOLUTION                |
| <b>Transport hazard class(es)</b> | 8                                      |
| <b>Packing group</b>              | II                                     |
| <b>EmS-No</b>                     | F-A, S-B                               |
| <b>Marine pollutant</b>           | NP                                     |
| <b>Description</b>                | UN2796, SULPHURIC ACID SOLUTION, 8, II |

## 15. Regulatory information

**International Inventories**

Contact supplier for inventory compliance status

**US Federal Regulations****SARA 313**

Section 313 of Title III of the Superfund Amendments and Reauthorization Act of 1986 (SARA). This product contains a chemical or chemicals which are subject to the reporting requirements of the Act and Title 40 of the Code of Federal Regulations, Part 372.

| Chemical name             | SARA 313 - Threshold Values % |
|---------------------------|-------------------------------|
| Sulfuric acid - 7664-93-9 | 1.0                           |

**SARA 311/312 Hazard Categories**

Should this product meet EPCRA 311/312 Tier reporting criteria at 40 CFR 370, refer to Section 2 of this SDS for appropriate classifications.

**CWA (Clean Water Act)**

This product contains the following substances which are regulated pollutants pursuant to the Clean Water Act (40 CFR 122.21 and 40 CFR 122.42).

| Chemical name              | CWA - Reportable Quantities | CWA - Toxic Pollutants | CWA - Priority Pollutants | CWA - Hazardous Substances |
|----------------------------|-----------------------------|------------------------|---------------------------|----------------------------|
| Sulfuric acid<br>7664-93-9 | 1000 lb                     | -                      | -                         | X                          |

**CERCLA**

This material, as supplied, contains one or more substances regulated as a hazardous substance under the Comprehensive Environmental Response Compensation and Liability Act (CERCLA) (40 CFR 302).

| Chemical name              | Hazardous Substances RQs | Extremely Hazardous Substances RQs | Reportable Quantity (RQ)                  |
|----------------------------|--------------------------|------------------------------------|-------------------------------------------|
| Sulfuric acid<br>7664-93-9 | 1000 lb                  | 1000 lb                            | RQ 1000 lb final RQ<br>RQ 454 kg final RQ |

**US State Regulations****California Proposition 65**

This product contains the following Proposition 65 chemicals:.

| Chemical name             | California Proposition 65 |
|---------------------------|---------------------------|
| Sulfuric acid - 7664-93-9 | Carcinogen                |

**U.S. State Right-to-Know Regulations**

| Chemical name      | New Jersey | Massachusetts | Pennsylvania |
|--------------------|------------|---------------|--------------|
| Water<br>7732-18-5 | -          | -             | X            |
| Sulfuric acid      | X          | X             | X            |

|           |  |  |  |
|-----------|--|--|--|
| 7664-93-9 |  |  |  |
|-----------|--|--|--|

**U.S. EPA Label Information**

EPA Pesticide Registration Number Not applicable

**16. Other information**

|             |                         |                       |                           |                              |
|-------------|-------------------------|-----------------------|---------------------------|------------------------------|
| <b>NFPA</b> | <b>Health hazards</b> 3 | <b>Flammability</b> 0 | <b>Instability</b> 0      | <b>Special hazards</b> -     |
| <b>HMIS</b> | <b>Health hazards</b> 3 | <b>Flammability</b> 0 | <b>Physical hazards</b> 0 | <b>Personal protection</b> X |

**Key or legend to abbreviations and acronyms used in the safety data sheet****Legend Section 8: EXPOSURE CONTROLS/PERSONAL PROTECTION**

|         |                             |      |                                  |
|---------|-----------------------------|------|----------------------------------|
| TWA     | TWA (time-weighted average) | STEL | STEL (Short Term Exposure Limit) |
| Ceiling | Maximum limit value         | *    | Skin designation                 |

**Key literature references and sources for data used to compile the SDS**

Agency for Toxic Substances and Disease Registry (ATSDR)  
 U.S. Environmental Protection Agency ChemView Database  
 European Food Safety Authority (EFSA)  
 EPA (Environmental Protection Agency)  
 Acute Exposure Guideline Level(s) (AEGLS)  
 U.S. Environmental Protection Agency Federal Insecticide, Fungicide, and Rodenticide Act  
 U.S. Environmental Protection Agency High Production Volume Chemicals  
 Food Research Journal  
 Hazardous Substance Database  
 International Uniform Chemical Information Database (IUCLID)  
 National Institute of Technology and Evaluation (NITE)  
 Australia National Industrial Chemicals Notification and Assessment Scheme (NICNAS)  
 NIOSH (National Institute for Occupational Safety and Health)  
 National Library of Medicine's ChemID Plus (NLM CIP)  
 National Library of Medicine's PubMed database (NLM PUBMED)  
 National Toxicology Program (NTP)  
 New Zealand's Chemical Classification and Information Database (CCID)  
 Organization for Economic Co-operation and Development Environment, Health, and Safety Publications  
 Organization for Economic Co-operation and Development High Production Volume Chemicals Program  
 Organization for Economic Co-operation and Development Screening Information Data Set  
 World Health Organization

Revision date 02-Mar-2022

Revision Note Significant changes throughout SDS. Review all sections.

**Disclaimer**

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text.

**End of Safety Data Sheet**



# SAFETY DATA SHEET

Revision date 02-Mar-2022

Revision Number 1

## 1. Identification

### Product identifier

**Product Name** R2 - 20 x Conc. Washing Solution, 70 ml

### Other means of identification

**Catalog Number(s)** 7361A, 7360S, 7360Z

### Recommended use of the chemical and restrictions on use

**Recommended use** In vitro diagnostic  
Restricted to professional users

**Restrictions on use** No information available

### Details of the supplier of the safety data sheet

#### **Corporate Headquarters**

Bio-Rad Laboratories Inc.  
1000 Alfred Nobel Drive  
Hercules, CA 94547  
USA

#### **Manufacturer Address**

Bio-Rad  
3 boulevard Raymond Poincaré  
92430 Marnes-la-Coquette  
France  
e-mail: fds-msds.fr@bio-rad.com

#### **Legal Entity / Contact Address**

Bio-Rad Eugenia 197 piso 10 -A  
Col. Narvarte Del. Benito Juárez  
CP. 03020, CDMX  
México

**Technical Service** +5255548870 ext. 1031 & 1083  
cts\_cdg\_mexico@bio-rad.com

### Emergency telephone number

**24 Hour Emergency Phone Number** CHEMTREC México: 01-800-681-9531

## 2. Hazard(s) identification

### Classification

Not classified.

### Label elements

#### **Hazard statements**

Not classified.

### Other information

Not applicable

## 3. Composition/information on ingredients

### Substance

Not applicable.

**Mixture**

| Chemical name     | CAS No    | Weight-%    |
|-------------------|-----------|-------------|
| Sodium chloride   | 7647-14-5 | 20 - 35     |
| Hydrochloric acid | 7647-01-0 | 0.3 - 0.999 |

**4. First-aid measures****Description of first aid measures**

|                |                                                                                                                      |
|----------------|----------------------------------------------------------------------------------------------------------------------|
| General advice | No hazards which require special first aid measures.                                                                 |
| Inhalation     | Remove to fresh air.                                                                                                 |
| Eye contact    | Rinse thoroughly with plenty of water for at least 15 minutes, lifting lower and upper eyelids. Consult a physician. |
| Skin contact   | Wash skin with soap and water.                                                                                       |
| Ingestion      | Rinse mouth thoroughly with water.                                                                                   |

**Most important symptoms and effects, both acute and delayed**

|          |                          |
|----------|--------------------------|
| Symptoms | No information available |
|----------|--------------------------|

**Indication of any immediate medical attention and special treatment needed**

|                    |                        |
|--------------------|------------------------|
| Note to physicians | Treat symptomatically. |
|--------------------|------------------------|

**5. Fire-fighting measures**

|                                              |                                                                                                                                    |
|----------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------|
| Suitable Extinguishing Media                 | Use extinguishing measures that are appropriate to local circumstances and the surrounding environment.                            |
| Unsuitable extinguishing media               | No information available.                                                                                                          |
| Specific hazards arising from the chemical   | None known.                                                                                                                        |
| Explosion data                               |                                                                                                                                    |
| Sensitivity to mechanical impact             | None.                                                                                                                              |
| Sensitivity to static discharge              | None.                                                                                                                              |
| Special protective actions for fire-fighters | Firefighters should wear self-contained breathing apparatus and full firefighting turnout gear. Use personal protection equipment. |

**6. Accidental release measures****Personal precautions, protective equipment and emergency procedures**

|                      |                                     |
|----------------------|-------------------------------------|
| Personal precautions | See section 8 for more information. |
|----------------------|-------------------------------------|

**Environmental precautions**

|                           |                                                       |
|---------------------------|-------------------------------------------------------|
| Environmental precautions | See Section 12 for additional Ecological Information. |
|---------------------------|-------------------------------------------------------|

**Methods and material for containment and cleaning up**

|                         |                                                       |
|-------------------------|-------------------------------------------------------|
| Methods for containment | Prevent further leakage or spillage if safe to do so. |
| Methods for cleaning up | Pick up and transfer to properly labeled containers.  |

**Prevention of secondary hazards** Clean contaminated objects and areas thoroughly observing environmental regulations.

## 7. Handling and storage

### Precautions for safe handling

**Advice on safe handling** Handle in accordance with good industrial hygiene and safety practice.

### Conditions for safe storage, including any incompatibilities

**Storage Conditions** Store according to product and label instructions.

## 8. Exposure controls/personal protection

### Control parameters

**Exposure Limits** NOM-010-STPS-2014.

| Chemical name                  | TWA | STEL | Ceiling Limit Value |
|--------------------------------|-----|------|---------------------|
| Hydrochloric acid<br>7647-01-0 | -   | -    | 2 ppm               |

### Appropriate engineering controls

**Engineering controls** Showers  
Eyewash stations  
Ventilation systems.

### Individual protection measures, such as personal protective equipment

**Eye/face protection** Wear safety glasses with side shields (or goggles).

**Hand protection** Wear suitable gloves.

**Skin and body protection** Wear suitable protective clothing.

**Respiratory protection** No protective equipment is needed under normal use conditions. If exposure limits are exceeded or irritation is experienced, ventilation and evacuation may be required.

**General hygiene considerations** Handle in accordance with good industrial hygiene and safety practice.

## 9. Physical and chemical properties

### Information on basic physical and chemical properties

**Physical state** Liquid  
**Appearance** aqueous solution  
**Color** colorless  
**Odor** Odorless  
**Odor threshold** No information available

| <u>Property</u>                        | <u>Values</u>     | <u>Remarks • Method</u> |
|----------------------------------------|-------------------|-------------------------|
| pH                                     | 7.4               |                         |
| Melting point / freezing point         | No data available | None known              |
| Boiling point / boiling range          | 100 °C / 212 °F   |                         |
| Flash point                            | No data available | None known              |
| Evaporation rate                       | No data available | None known              |
| Flammability (solid, gas)              | No data available | None known              |
| Flammability Limit in Air              |                   | None known              |
| Upper flammability or explosive limits | No data available |                         |
| Lower flammability or explosive limits | No data available |                         |

|                              |                   |            |
|------------------------------|-------------------|------------|
| Vapor pressure               | No data available | None known |
| Vapor density                | No data available | None known |
| Relative density             | No data available | None known |
| Water solubility             | Miscible in water |            |
| Solubility in other solvents | No data available | None known |
| Partition coefficient        | No data available | None known |
| Autoignition temperature     | No data available | None known |
| Decomposition temperature    | No data available | None known |
| Kinematic viscosity          | No data available | None known |
| Dynamic viscosity            | No data available | None known |

**Other information**

|                      |                 |
|----------------------|-----------------|
| Oxidizing properties | Not applicable. |
| Explosive properties | Not applicable. |
| Softening point      | Not applicable  |
| Molecular weight     | Not applicable  |
| VOC Content (%)      | Not applicable  |

**10. Stability and reactivity**

|                                    |                                           |
|------------------------------------|-------------------------------------------|
| Reactivity                         | No information available.                 |
| Chemical stability                 | Stable under normal conditions.           |
| Possibility of hazardous reactions | None under normal processing.             |
| Conditions to avoid                | None known based on information supplied. |
| Incompatible materials             | None known based on information supplied. |
| Hazardous decomposition products   | None known based on information supplied. |

**11. Toxicological information****Information on likely routes of exposure**

|              |                                                                   |
|--------------|-------------------------------------------------------------------|
| Inhalation   | Specific test data for the substance or mixture is not available. |
| Eye contact  | Specific test data for the substance or mixture is not available. |
| Skin contact | Specific test data for the substance or mixture is not available. |
| Ingestion    | Specific test data for the substance or mixture is not available. |

**Symptoms related to the physical, chemical and toxicological characteristics**

|          |                           |
|----------|---------------------------|
| Symptoms | No information available. |
|----------|---------------------------|

**Acute toxicity****Numerical measures of toxicity**

The following values are calculated based on chapter 3.1 of the GHS document

ATEmix (oral) 11,155.50 mg/kg

**Component Information**

| Chemical name                  | Oral LD50               | Dermal LD50              | Inhalation LC50         |
|--------------------------------|-------------------------|--------------------------|-------------------------|
| Sodium chloride<br>7647-14-5   | = 3 g/kg ( Rat )        | > 10000 mg/kg ( Rabbit ) | > 42 mg/L ( Rat ) 1 h   |
| Hydrochloric acid<br>7647-01-0 | 238 - 277 mg/kg ( Rat ) | > 5010 mg/kg ( Rabbit )  | = 1.68 mg/L ( Rat ) 1 h |

**Delayed and immediate effects as well as chronic effects from short and long-term exposure**

|                                          |                                                                   |
|------------------------------------------|-------------------------------------------------------------------|
| <b>Interactive effects</b>               | Based on available data, the classification criteria are not met. |
| <b>Skin corrosion/irritation</b>         | Based on available data, the classification criteria are not met. |
| <b>Serious eye damage/eye irritation</b> | Based on available data, the classification criteria are not met. |
| <b>Respiratory or skin sensitization</b> | Based on available data, the classification criteria are not met. |
| <b>Germ cell mutagenicity</b>            | Based on available data, the classification criteria are not met. |

**Carcinogenicity**

The table below indicates whether each agency has listed any ingredient as a carcinogen.

| Chemical name                  | ACGIH | IARC    | NTP | Mexico |
|--------------------------------|-------|---------|-----|--------|
| Hydrochloric acid<br>7647-01-0 | -     | Group 3 | -   | -      |

**Legend**

**IARC (International Agency for Research on Cancer)**

Group 3 - Not Classifiable as to Carcinogenicity in Humans

|                                 |                                                                   |
|---------------------------------|-------------------------------------------------------------------|
| <b>Reproductive toxicity</b>    | Based on available data, the classification criteria are not met. |
| <b>STOT - single exposure</b>   | Based on available data, the classification criteria are not met. |
| <b>STOT - repeated exposure</b> | Based on available data, the classification criteria are not met. |
| <b>Target organ effects</b>     | Respiratory system, Eyes, Skin.                                   |
| <b>Aspiration hazard</b>        | Based on available data, the classification criteria are not met. |
| <b>Other information</b>        | No information available.                                         |

**12. Ecological information****Ecotoxicity**

| Chemical name                | Algae/aquatic plants | Fish                                                                                                                                                                                                                                                                                                      | Toxicity to microorganisms | Crustacea                                                                            |
|------------------------------|----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------|--------------------------------------------------------------------------------------|
| Sodium chloride<br>7647-14-5 | -                    | LC50: 4747 - 7824mg/L (96h, Oncorhynchus mykiss)<br>LC50: 5560 - 6080mg/L (96h, Lepomis macrochirus)<br>LC50: 6020 - 7070mg/L (96h, Pimephales promelas)<br>LC50: 6420 - 6700mg/L (96h, Pimephales promelas)<br>LC50: =12946mg/L (96h, Lepomis macrochirus)<br>LC50: =7050mg/L (96h, Pimephales promelas) | -                          | EC50: 340.7 - 469.2mg/L (48h, Daphnia magna)<br>EC50: =1000mg/L (48h, Daphnia magna) |

**Persistence and degradability** No information available.

**Bioaccumulation** No information available.

Other adverse effects No information available.

### 13. Disposal considerations

#### Waste treatment methods

Waste from residues/unused products Dispose of in accordance with local regulations. Dispose of waste in accordance with environmental legislation.

Contaminated packaging Do not reuse empty containers.

### 14. Transport information

MEX Not regulated

Transport in bulk according to Annex II of MARPOL 73/78 and the IBC Code No information available

TDG Not regulated

DOT Not regulated

IATA Not regulated

IMDG Not regulated

### 15. Regulatory information

#### Safety, health and environmental regulations/legislation specific for the substance or mixture

#### International Regulations

The Montreal Protocol on Substances that Deplete the Ozone Layer Not applicable

The Stockholm Convention on Persistent Organic Pollutants Not applicable

The Rotterdam Convention Not applicable

#### International Inventories

Contact supplier for inventory compliance status

### 16. Other information

|             |                  |                |                    |                                    |
|-------------|------------------|----------------|--------------------|------------------------------------|
| <u>NFPA</u> | Health hazards 0 | Flammability 0 | Instability 0      | Physical and chemical properties - |
| <u>HMIS</u> | Health hazards 0 | Flammability 0 | Physical hazards 0 | Personal protection X              |

#### Key or legend to abbreviations and acronyms used in the safety data sheet

#### Legend Section 8: EXPOSURE CONTROLS/PERSONAL PROTECTION

|         |                             |      |                                  |
|---------|-----------------------------|------|----------------------------------|
| TWA     | TWA (time-weighted average) | STEL | STEL (Short Term Exposure Limit) |
| Ceiling | Maximum limit value         | *    | Skin designation                 |

#### Key literature references and sources for data used to compile the SDS

Agency for Toxic Substances and Disease Registry (ATSDR)  
U.S. Environmental Protection Agency ChemView Database  
European Food Safety Authority (EFSA)



EPA (Environmental Protection Agency)  
Acute Exposure Guideline Level(s) (AEGL(s))  
U.S. Environmental Protection Agency Federal Insecticide, Fungicide, and Rodenticide Act  
U.S. Environmental Protection Agency High Production Volume Chemicals  
Food Research Journal  
Hazardous Substance Database  
International Uniform Chemical Information Database (IUCLID)  
Japan GHS Classification  
Australia National Industrial Chemicals Notification and Assessment Scheme (NICNAS)  
NIOSH (National Institute for Occupational Safety and Health)  
National Library of Medicine's ChemID Plus (NLM CIP)  
National Library of Medicine's PubMed database (NLM PUBMED)  
National Toxicology Program (NTP)  
New Zealand's Chemical Classification and Information Database (CCID)  
Organization for Economic Co-operation and Development Environment, Health, and Safety Publications  
Organization for Economic Co-operation and Development High Production Volume Chemicals Program  
Organization for Economic Co-operation and Development Screening Information Data Set  
RTECS (Registry of Toxic Effects of Chemical Substances)  
World Health Organization

**Prepared By** Bio-Rad Laboratories, Environmental Health and Safety.

**Revision date** 02-Mar-2022

**Revision Note** Significant changes throughout SDS. Review all sections.

**NOM-018-STPS-2015**

The information is believed to be accurate, but it is not exhaustive and must be used only as guidance. It is based on the current state of knowledge of the chemical substance or mixture and is applicable to the appropriate safety precautions for the product.

**Disclaimer**

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text.

**End of Safety Data Sheet**

# SAFETY DATA SHEET



Revision date 03-Mar-2022

Revision Number 1

## 1. Identification

### Product identifier

**Product Name** R9 - Chromogen: TMB Solution (11x), (5 mL)

### Other means of identification

**Catalog Number(s)** 7436L, 7436H

### Recommended use of the chemical and restrictions on use

**Recommended use** Restricted to professional users In vitro diagnostic

### Details of the supplier of the safety data sheet

#### Corporate Headquarters

Bio-Rad Laboratories Inc.  
1000 Alfred Nobel Drive  
Hercules, CA 94547  
USA

#### Manufacturer Address

Bio-Rad  
3 boulevard Raymond Poincaré  
92430 Marnes-la-Coquette  
France  
e-mail: fds-msds.fr@bio-rad.com

#### Legal Entity / Contact Address

Bio-Rad Laboratories  
Clinical Diagnostics  
4000 Alfred Nobel Drive  
Hercules, California 94547  
USA

**Technical Service** 1-800-224-6723  
TechSupport.USSD@bio-rad.com

### Emergency telephone number

**24 Hour Emergency Phone Number** CHEMTREC USA: 1 (800) 424-9300

## 2. Hazard(s) identification

### Classification

|                                   |            |
|-----------------------------------|------------|
| Skin corrosion/irritation         | Category 1 |
| Serious eye damage/eye irritation | Category 1 |

### Hazards not otherwise classified (HNOC)

Not applicable

### Label elements

**Danger**

#### Hazard statements

Causes severe skin burns and eye damage



**Appearance** Liquid

**Physical state** Liquid

**Odor** Low

**Precautionary Statements - Prevention**

Do not breathe dusts or mists

Wash face, hands and any exposed skin thoroughly after handling

Wear protective gloves/protective clothing/eye protection/face protection

**Precautionary Statements - Response**

Immediately call a POISON CENTER or doctor

IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing

Immediately call a POISON CENTER or doctor

IF ON SKIN (or hair): Take off immediately all contaminated clothing. Rinse skin with water/ shower

Wash contaminated clothing before reuse

IF INHALED: Remove person to fresh air and keep comfortable for breathing

Immediately call a POISON CENTER or doctor

IF SWALLOWED: Rinse mouth. Do NOT induce vomiting

**Precautionary Statements - Disposal**

Dispose of contents/container in accordance with local, regional, national, and international regulations as applicable

**Other information**

No information available.

### 3. Composition/information on ingredients

**Substance**

Not applicable.

**Mixture**

| Chemical name     | CAS No    | Weight-%    | Trade secret |
|-------------------|-----------|-------------|--------------|
| Hydrochloric acid | 7647-01-0 | 0.3 - 0.999 | *            |

\*The exact percentage (concentration) of composition has been withheld as a trade secret.

### 4. First-aid measures

**Description of first aid measures****General advice**

Immediate medical attention is required. Show this safety data sheet to the doctor in attendance.

**Inhalation**

Remove to fresh air. If breathing has stopped, give artificial respiration. Get medical attention immediately. Do not use mouth-to-mouth method if victim ingested or inhaled the substance; give artificial respiration with the aid of a pocket mask equipped with a one-way valve or other proper respiratory medical device. If breathing is difficult, (trained personnel should) give oxygen. Delayed pulmonary edema may occur. Get immediate medical advice/attention.

**Eye contact**

Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Keep eye wide open while rinsing. Do not rub affected area. Remove contact lenses, if present and easy to do. Continue rinsing. Get immediate medical advice/attention.

**Skin contact**

Wash off immediately with soap and plenty of water while removing all contaminated clothes and shoes. Get immediate medical advice/attention.

**Ingestion**

Do NOT induce vomiting. Rinse mouth. Never give anything by mouth to an unconscious person. Get immediate medical advice/attention.

**Self-protection of the first aider**

Ensure that medical personnel are aware of the material(s) involved, take precautions to protect themselves and prevent spread of contamination. Avoid contact with skin, eyes or clothing. Avoid direct contact with skin. Use barrier to give mouth-to-mouth resuscitation.

Wear personal protective clothing (see section 8).

**Most important symptoms and effects, both acute and delayed**

**Symptoms** Burning sensation.

**Indication of any immediate medical attention and special treatment needed**

**Note to physicians** Product is a corrosive material. Use of gastric lavage or emesis is contraindicated. Possible perforation of stomach or esophagus should be investigated. Do not give chemical antidotes. Asphyxia from glottal edema may occur. Marked decrease in blood pressure may occur with moist rales, frothy sputum, and high pulse pressure.

## 5. Fire-fighting measures

**Suitable Extinguishing Media** Use extinguishing measures that are appropriate to local circumstances and the surrounding environment.

**Specific hazards arising from the chemical** The product causes burns of eyes, skin and mucous membranes. Thermal decomposition can lead to release of irritating gases and vapors.

**Explosion data**

**Sensitivity to mechanical impact** None.

**Sensitivity to static discharge** None.

**Special protective equipment and precautions for fire-fighters** Firefighters should wear self-contained breathing apparatus and full firefighting turnout gear. Use personal protection equipment.

## 6. Accidental release measures

**Personal precautions, protective equipment and emergency procedures**

**Personal precautions** Attention! Corrosive material. Avoid contact with skin, eyes or clothing. Ensure adequate ventilation. Use personal protective equipment as required. Evacuate personnel to safe areas. Keep people away from and upwind of spill/leak.

**Other information** Refer to protective measures listed in Sections 7 and 8.

**Methods and material for containment and cleaning up**

**Methods for containment** Prevent further leakage or spillage if safe to do so.

**Methods for cleaning up** Pick up and transfer to properly labeled containers.

## 7. Handling and storage

**Precautions for safe handling**

**Advice on safe handling** Handle in accordance with good industrial hygiene and safety practice. Avoid contact with skin, eyes or clothing. In case of insufficient ventilation, wear suitable respiratory equipment. Handle product only in closed system or provide appropriate exhaust ventilation. Do not eat, drink or smoke when using this product. Take off contaminated clothing and wash before reuse.

**Conditions for safe storage, including any incompatibilities**

**Storage Conditions** Keep containers tightly closed in a dry, cool and well-ventilated place. Protect from

moisture. Store locked up. Keep out of the reach of children. Store away from other materials. Store according to product and label instructions.

## 8. Exposure controls/personal protection

### Control parameters

#### Exposure Limits

| Chemical name                  | ACGIH TLV      | OSHA PEL                                                                                                             | NIOSH                                                          |
|--------------------------------|----------------|----------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------|
| Hydrochloric acid<br>7647-01-0 | Ceiling: 2 ppm | (vacated) Ceiling: 5 ppm<br>(vacated) Ceiling: 7 mg/m <sup>3</sup><br>Ceiling: 5 ppm<br>Ceiling: 7 mg/m <sup>3</sup> | IDLH: 50 ppm<br>Ceiling: 5 ppm<br>Ceiling: 7 mg/m <sup>3</sup> |

### Appropriate engineering controls

**Engineering controls**                      Showers  
                                                          Eyewash stations  
                                                          Ventilation systems.

### Individual protection measures, such as personal protective equipment

**Eye/face protection**                      Tight sealing safety goggles. Face protection shield.

**Hand protection**                              Wear suitable gloves. Impervious gloves.

**Skin and body protection**                      Wear suitable protective clothing. Long sleeved clothing. Chemical resistant apron.

**Respiratory protection**                      No protective equipment is needed under normal use conditions. If exposure limits are exceeded or irritation is experienced, ventilation and evacuation may be required.

**General hygiene considerations**                      Avoid contact with skin, eyes or clothing. Wear suitable gloves and eye/face protection. Do not eat, drink or smoke when using this product. Remove and wash contaminated clothing and gloves, including the inside, before re-use. Contaminated work clothing should not be allowed out of the workplace. Regular cleaning of equipment, work area and clothing is recommended. Wash hands before breaks and immediately after handling the product.

## 9. Physical and chemical properties

### Information on basic physical and chemical properties

**Physical state**                                      Liquid  
**Appearance**                                      Liquid  
**Color**                                                  pink  
**Odor**                                                  Low  
**Odor threshold**                                      No information available

| <u>Property</u>                        | <u>Values</u>     | <u>Remarks • Method</u> |
|----------------------------------------|-------------------|-------------------------|
| pH                                     |                   |                         |
| Melting point / freezing point         | No data available | None known              |
| Boiling point / boiling range          | No data available | None known              |
| Flash point                            | No data available | None known              |
| Evaporation rate                       | No data available | None known              |
| Flammability (solid, gas)              | No data available | None known              |
| Flammability Limit in Air              |                   | None known              |
| Upper flammability or explosive limits | No data available |                         |
| Lower flammability or explosive        | No data available |                         |

|                           |                   |            |
|---------------------------|-------------------|------------|
| <b>limits</b>             |                   |            |
| Vapor pressure            | No data available | None known |
| Vapor density             | No data available | None known |
| Relative density          | No data available | None known |
| Water solubility          | Miscible in water |            |
| Solubility(ies)           | No data available | None known |
| Partition coefficient     | No data available | None known |
| Autoignition temperature  | 363               | None known |
| Decomposition temperature |                   | None known |
| Kinematic viscosity       | No data available | None known |
| Dynamic viscosity         | No data available | None known |

**Other information**

|                      |                          |
|----------------------|--------------------------|
| Explosive properties | No information available |
| Oxidizing properties | No information available |
| Softening point      | No information available |
| Molecular weight     | No information available |
| VOC Content (%)      | No information available |
| Liquid Density       | No information available |
| Bulk density         | No information available |

## 10. Stability and reactivity

|                                    |                                                     |
|------------------------------------|-----------------------------------------------------|
| Reactivity                         | No information available.                           |
| Chemical stability                 | Stable under normal conditions.                     |
| Possibility of hazardous reactions | None under normal processing.                       |
| Conditions to avoid                | Exposure to air or moisture over prolonged periods. |
| Incompatible materials             | Acids. Bases. Oxidizing agent.                      |
| Hazardous decomposition products   | None known based on information supplied.           |

## 11. Toxicological information

**Information on likely routes of exposure****Product Information**

|                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|---------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Inhalation</b>   | Specific test data for the substance or mixture is not available. Corrosive by inhalation. (based on components). Inhalation of corrosive fumes/gases may cause coughing, choking, headache, dizziness, and weakness for several hours. Pulmonary edema may occur with tightness in the chest, shortness of breath, bluish skin, decreased blood pressure, and increased heart rate. Inhaled corrosive substances can lead to a toxic edema of the lungs. Pulmonary edema can be fatal.                                   |
| <b>Eye contact</b>  | Specific test data for the substance or mixture is not available. Causes serious eye damage. (based on components). Corrosive to the eyes and may cause severe damage including blindness. May cause irreversible damage to eyes.                                                                                                                                                                                                                                                                                         |
| <b>Skin contact</b> | Specific test data for the substance or mixture is not available. Corrosive. (based on components). Causes burns.                                                                                                                                                                                                                                                                                                                                                                                                         |
| <b>Ingestion</b>    | Specific test data for the substance or mixture is not available. Causes burns. (based on components). Ingestion causes burns of the upper digestive and respiratory tracts. May cause severe burning pain in the mouth and stomach with vomiting and diarrhea of dark blood. Blood pressure may decrease. Brownish or yellowish stains may be seen around the mouth. Swelling of the throat may cause shortness of breath and choking. May cause lung damage if swallowed. May be fatal if swallowed and enters airways. |

**Symptoms related to the physical, chemical and toxicological characteristics**

**Symptoms** Redness. Burning. May cause blindness. Coughing and/ or wheezing.

**Acute toxicity****Numerical measures of toxicity****Component Information**

| Chemical name                  | Oral LD50               | Dermal LD50             | Inhalation LC50         |
|--------------------------------|-------------------------|-------------------------|-------------------------|
| Hydrochloric acid<br>7647-01-0 | 238 - 277 mg/kg ( Rat ) | > 5010 mg/kg ( Rabbit ) | = 1.68 mg/L ( Rat ) 1 h |

**Delayed and immediate effects as well as chronic effects from short and long-term exposure**

**Skin corrosion/irritation** Classification based on data available for ingredients. Causes burns.

**Serious eye damage/eye irritation** Classification based on data available for ingredients. Risk of serious damage to eyes. Causes burns.

**Respiratory or skin sensitization** No information available.

**Germ cell mutagenicity** No information available.

**Carcinogenicity** No information available.

The table below indicates whether each agency has listed any ingredient as a carcinogen.

| Chemical name                  | ACGIH | IARC    | NTP | OSHA |
|--------------------------------|-------|---------|-----|------|
| Hydrochloric acid<br>7647-01-0 | -     | Group 3 | -   | X    |

**Legend**

**IARC (International Agency for Research on Cancer)**

Group 3 - Not Classifiable as to Carcinogenicity in Humans

**OSHA (Occupational Safety and Health Administration of the US Department of Labor)**

X - Present

**Reproductive toxicity** No information available.

**STOT - single exposure** No information available.

**STOT - repeated exposure** No information available.

**Aspiration hazard** No information available.

**Other adverse effects** No information available.

**Interactive effects** No information available.

## 12. Ecological information

**Ecotoxicity**

**Persistence and degradability** No information available.

**Bioaccumulation** There is no data for this product.

**Other adverse effects** No information available.

### 13. Disposal considerations

#### Waste treatment methods

**Waste from residues/unused products** Dispose of in accordance with local regulations. Dispose of waste in accordance with environmental legislation.

**Contaminated packaging** Do not reuse empty containers.

**US EPA Waste Number** D002.

### 14. Transport information

**DOT** Not regulated

**TDG** Not regulated

**MEX** Not regulated

**IATA** Not regulated

**IMDG** Not regulated

### 15. Regulatory information

**International Inventories** Contact supplier for inventory compliance status

#### US Federal Regulations

##### **SARA 313**

Section 313 of Title III of the Superfund Amendments and Reauthorization Act of 1986 (SARA). This product contains a chemical or chemicals which are subject to the reporting requirements of the Act and Title 40 of the Code of Federal Regulations, Part 372.

| Chemical name                 | SARA 313 - Threshold Values % |
|-------------------------------|-------------------------------|
| Hydrochloric acid - 7647-01-0 | 1.0                           |

##### **SARA 311/312 Hazard Categories**

Should this product meet EPCRA 311/312 Tier reporting criteria at 40 CFR 370, refer to Section 2 of this SDS for appropriate classifications.

##### **CWA (Clean Water Act)**

This product contains the following substances which are regulated pollutants pursuant to the Clean Water Act (40 CFR 122.21 and 40 CFR 122.42).

| Chemical name     | CWA - Reportable Quantities | CWA - Toxic Pollutants | CWA - Priority Pollutants | CWA - Hazardous Substances |
|-------------------|-----------------------------|------------------------|---------------------------|----------------------------|
| Hydrochloric acid | 5000 lb                     | -                      | -                         | X                          |



**R9 - Chromogen: TMB Solution (11x), (5 mL)**

Revision date 03-Mar-2022

|           |  |  |  |  |
|-----------|--|--|--|--|
| 7647-01-0 |  |  |  |  |
|-----------|--|--|--|--|

**CERCLA**

This material, as supplied, contains one or more substances regulated as a hazardous substance under the Comprehensive Environmental Response Compensation and Liability Act (CERCLA) (40 CFR 302).

| Chemical name                  | Hazardous Substances RQs | Extremely Hazardous Substances RQs | Reportable Quantity (RQ)                   |
|--------------------------------|--------------------------|------------------------------------|--------------------------------------------|
| Hydrochloric acid<br>7647-01-0 | 5000 lb                  | 5000 lb                            | RQ 5000 lb final RQ<br>RQ 2270 kg final RQ |

**US State Regulations****California Proposition 65**

This product contains the following Proposition 65 chemicals:.

| Chemical name           | California Proposition 65   |
|-------------------------|-----------------------------|
| Ethyl alcohol - 64-17-5 | Carcinogen<br>Developmental |

**U.S. State Right-to-Know Regulations**

| Chemical name                  | New Jersey | Massachusetts | Pennsylvania |
|--------------------------------|------------|---------------|--------------|
| Water<br>7732-18-5             | -          | -             | X            |
| Hydrochloric acid<br>7647-01-0 | X          | X             | X            |
| Ethyl alcohol<br>64-17-5       | X          | X             | X            |

**U.S. EPA Label Information**

**EPA Pesticide Registration Number** Not applicable

**16. Other information**

|             |                         |                       |                           |                              |
|-------------|-------------------------|-----------------------|---------------------------|------------------------------|
| <b>NFPA</b> | <b>Health hazards</b> 3 | <b>Flammability</b> 1 | <b>Instability</b> 0      | <b>Special hazards</b> -     |
| <b>HMIS</b> | <b>Health hazards</b> 3 | <b>Flammability</b> 1 | <b>Physical hazards</b> 0 | <b>Personal protection</b> X |

**Key or legend to abbreviations and acronyms used in the safety data sheet****Legend Section 8: EXPOSURE CONTROLS/PERSONAL PROTECTION**

|         |                             |      |                                  |
|---------|-----------------------------|------|----------------------------------|
| TWA     | TWA (time-weighted average) | STEL | STEL (Short Term Exposure Limit) |
| Ceiling | Maximum limit value         | *    | Skin designation                 |

**Key literature references and sources for data used to compile the SDS**

Agency for Toxic Substances and Disease Registry (ATSDR)  
 U.S. Environmental Protection Agency ChemView Database  
 European Food Safety Authority (EFSA)  
 EPA (Environmental Protection Agency)  
 Acute Exposure Guideline Level(s) (AEGl(s))  
 U.S. Environmental Protection Agency Federal Insecticide, Fungicide, and Rodenticide Act  
 U.S. Environmental Protection Agency High Production Volume Chemicals  
 Food Research Journal  
 Hazardous Substance Database  
 International Uniform Chemical Information Database (IUCLID)  
 National Institute of Technology and Evaluation (NITE)  
 Australia National Industrial Chemicals Notification and Assessment Scheme (NICNAS)  
 NIOSH (National Institute for Occupational Safety and Health)  
 National Library of Medicine's ChemID Plus (NLM CIP)  
 National Library of Medicine's PubMed database (NLM PUBMED)  
 National Toxicology Program (NTP)

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New Zealand's Chemical Classification and Information Database (CCID)  
Organization for Economic Co-operation and Development Environment, Health, and Safety Publications  
Organization for Economic Co-operation and Development High Production Volume Chemicals Program  
Organization for Economic Co-operation and Development Screening Information Data Set  
World Health Organization

**Revision date** 03-Mar-2022

**Revision Note** Significant changes throughout SDS. Review all sections.

**Disclaimer**

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text.

**End of Safety Data Sheet**

# SAFETY DATA SHEET



Revision date 03-Mar-2022

Revision Number 1

## 1. Identification

### Product identifier

**Product Name** R1 - Microplate (12 strips x 8 wells)

### Other means of identification

**Catalog Number(s)** 7256B

### Recommended use of the chemical and restrictions on use

**Recommended use** Restricted to professional users In vitro diagnostic

### Details of the supplier of the safety data sheet

#### Corporate Headquarters

Bio-Rad Laboratories Inc.  
1000 Alfred Nobel Drive  
Hercules, CA 94547  
USA

#### Manufacturer Address

Bio-Rad  
3 boulevard Raymond Poincaré  
92430 Marnes-la-Coquette  
France  
e-mail: fds-msds.fr@bio-rad.com

#### Legal Entity / Contact Address

Bio-Rad Laboratories  
Clinical Diagnostics  
4000 Alfred Nobel Drive  
Hercules, California 94547  
USA

**Technical Service** 1-800-224-6723  
TechSupport.USSD@bio-rad.com

### Emergency telephone number

**24 Hour Emergency Phone Number** CHEMTREC USA: 1 (800) 424-9300

## 2. Hazard(s) identification

### Classification

This chemical is not considered hazardous by the 2012 OSHA Hazard Communication Standard (29 CFR 1910.1200)

### Hazards not otherwise classified (HNOC)

Not applicable

### Label elements

#### Hazard statements

This chemical is not considered hazardous by the 2012 OSHA Hazard Communication Standard (29 CFR 1910.1200)

**Appearance** solid

**Physical state** Solid

**Odor** Odorless

### Other information

No information available.

### 3. Composition/information on ingredients

#### Substance

The product contains no substances which at their given concentration, are considered to be hazardous to health

\*The exact percentage (concentration) of composition has been withheld as a trade secret.

### 4. First-aid measures

#### Description of first aid measures

|              |                                                                                                                      |
|--------------|----------------------------------------------------------------------------------------------------------------------|
| Inhalation   | Remove to fresh air.                                                                                                 |
| Eye contact  | Rinse thoroughly with plenty of water for at least 15 minutes, lifting lower and upper eyelids. Consult a physician. |
| Skin contact | Wash skin with soap and water.                                                                                       |
| Ingestion    | Rinse mouth.                                                                                                         |

#### Most important symptoms and effects, both acute and delayed

|          |                           |
|----------|---------------------------|
| Symptoms | No information available. |
|----------|---------------------------|

#### Indication of any immediate medical attention and special treatment needed

|                    |                        |
|--------------------|------------------------|
| Note to physicians | Treat symptomatically. |
|--------------------|------------------------|

### 5. Fire-fighting measures

|                                                                |                                                                                                                                    |
|----------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------|
| Suitable Extinguishing Media                                   | Use extinguishing measures that are appropriate to local circumstances and the surrounding environment.                            |
| Specific hazards arising from the chemical                     | No information available.                                                                                                          |
| Explosion data                                                 |                                                                                                                                    |
| Sensitivity to mechanical impact                               | None.                                                                                                                              |
| Sensitivity to static discharge                                | None.                                                                                                                              |
| Special protective equipment and precautions for fire-fighters | Firefighters should wear self-contained breathing apparatus and full firefighting turnout gear. Use personal protection equipment. |

### 6. Accidental release measures

#### Personal precautions, protective equipment and emergency procedures

|                      |                              |
|----------------------|------------------------------|
| Personal precautions | Ensure adequate ventilation. |
|----------------------|------------------------------|

#### Methods and material for containment and cleaning up

|                         |                                                       |
|-------------------------|-------------------------------------------------------|
| Methods for containment | Prevent further leakage or spillage if safe to do so. |
|-------------------------|-------------------------------------------------------|

|                         |                                                      |
|-------------------------|------------------------------------------------------|
| Methods for cleaning up | Pick up and transfer to properly labeled containers. |
|-------------------------|------------------------------------------------------|

## 7. Handling and storage

### Precautions for safe handling

|                         |                                                                        |
|-------------------------|------------------------------------------------------------------------|
| Advice on safe handling | Handle in accordance with good industrial hygiene and safety practice. |
|-------------------------|------------------------------------------------------------------------|

### Conditions for safe storage, including any incompatibilities

|                    |                                                                   |
|--------------------|-------------------------------------------------------------------|
| Storage Conditions | Keep container tightly closed in a dry and well-ventilated place. |
|--------------------|-------------------------------------------------------------------|

## 8. Exposure controls/personal protection

### Control parameters

|                 |                                                                                                                                                             |
|-----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Exposure Limits | This product, as supplied, does not contain any hazardous materials with occupational exposure limits established by the region specific regulatory bodies. |
|-----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|

### Appropriate engineering controls

|                      |                                                     |
|----------------------|-----------------------------------------------------|
| Engineering controls | Showers<br>Eyewash stations<br>Ventilation systems. |
|----------------------|-----------------------------------------------------|

### Individual protection measures, such as personal protective equipment

|                     |                                           |
|---------------------|-------------------------------------------|
| Eye/face protection | No special protective equipment required. |
|---------------------|-------------------------------------------|

|                          |                                           |
|--------------------------|-------------------------------------------|
| Skin and body protection | No special protective equipment required. |
|--------------------------|-------------------------------------------|

|                        |                                                                                                                                                                          |
|------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Respiratory protection | No protective equipment is needed under normal use conditions. If exposure limits are exceeded or irritation is experienced, ventilation and evacuation may be required. |
|------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

|                                |                                                                        |
|--------------------------------|------------------------------------------------------------------------|
| General hygiene considerations | Handle in accordance with good industrial hygiene and safety practice. |
|--------------------------------|------------------------------------------------------------------------|

## 9. Physical and chemical properties

### Information on basic physical and chemical properties

|                |                          |
|----------------|--------------------------|
| Physical state | Solid                    |
| Appearance     | solid                    |
| Color          | colorless                |
| Odor           | Odorless                 |
| Odor threshold | No information available |

| <u>Property</u>                | <u>Values</u>     | <u>Remarks • Method</u> |
|--------------------------------|-------------------|-------------------------|
| pH                             |                   | None known              |
| Melting point / freezing point | No data available | None known              |
| Boiling point / boiling range  | No data available | None known              |
| Flash point                    | No data available | None known              |
| Evaporation rate               | No data available | None known              |
| Flammability (solid, gas)      | No data available | None known              |
| Flammability Limit in Air      |                   | None known              |

|                                        |                          |            |
|----------------------------------------|--------------------------|------------|
| Upper flammability or explosive limits | No data available        |            |
| Lower flammability or explosive limits | No data available        |            |
| Vapor pressure                         | No data available        | None known |
| Vapor density                          | No data available        | None known |
| Relative density                       | No data available        | None known |
| Water solubility                       | Insoluble in water       |            |
| Solubility(ies)                        | No data available        | None known |
| Partition coefficient                  | No data available        | None known |
| Autoignition temperature               | No data available        | None known |
| Decomposition temperature              |                          | None known |
| Kinematic viscosity                    | No data available        | None known |
| Dynamic viscosity                      | No data available        | None known |
| <b><u>Other information</u></b>        |                          |            |
| Explosive properties                   | No information available |            |
| Oxidizing properties                   | No information available |            |
| Softening point                        | No information available |            |
| Molecular weight                       | No information available |            |
| VOC Content (%)                        | No information available |            |
| Liquid Density                         | No information available |            |
| Bulk density                           | No information available |            |

## 10. Stability and reactivity

|                                    |                                           |
|------------------------------------|-------------------------------------------|
| Reactivity                         | No information available.                 |
| Chemical stability                 | Stable under normal conditions.           |
| Possibility of hazardous reactions | None under normal processing.             |
| Conditions to avoid                | None known based on information supplied. |
| Incompatible materials             | None known based on information supplied. |
| Hazardous decomposition products   | None known based on information supplied. |

## 11. Toxicological information

### Information on likely routes of exposure

|              |                                                                   |
|--------------|-------------------------------------------------------------------|
| Inhalation   | Specific test data for the substance or mixture is not available. |
| Eye contact  | Specific test data for the substance or mixture is not available. |
| Skin contact | Specific test data for the substance or mixture is not available. |
| Ingestion    | Specific test data for the substance or mixture is not available. |

### Symptoms related to the physical, chemical and toxicological characteristics

|          |                           |
|----------|---------------------------|
| Symptoms | No information available. |
|----------|---------------------------|

### Acute toxicity

|                                |                          |
|--------------------------------|--------------------------|
| Numerical measures of toxicity | No information available |
|--------------------------------|--------------------------|

**Delayed and immediate effects as well as chronic effects from short and long-term exposure**

|                                          |                           |
|------------------------------------------|---------------------------|
| <b>Skin corrosion/irritation</b>         | No information available. |
| <b>Serious eye damage/eye irritation</b> | No information available. |
| <b>Respiratory or skin sensitization</b> | No information available. |
| <b>Germ cell mutagenicity</b>            | No information available. |
| <b>Carcinogenicity</b>                   | No information available. |
| <b>Reproductive toxicity</b>             | No information available. |
| <b>STOT - single exposure</b>            | No information available. |
| <b>STOT - repeated exposure</b>          | No information available. |
| <b>Aspiration hazard</b>                 | No information available. |
| <b>Other adverse effects</b>             | No information available. |
| <b>Interactive effects</b>               | No information available. |

**12. Ecological information**

|                                      |                                                                           |
|--------------------------------------|---------------------------------------------------------------------------|
| <b>Ecotoxicity</b>                   | The environmental impact of this product has not been fully investigated. |
| <b>Persistence and degradability</b> | No information available.                                                 |
| <b>Bioaccumulation</b>               | There is no data for this product.                                        |
| <b>Other adverse effects</b>         | No information available.                                                 |

**13. Disposal considerations****Waste treatment methods**

|                                            |                                                                                                                 |
|--------------------------------------------|-----------------------------------------------------------------------------------------------------------------|
| <b>Waste from residues/unused products</b> | Dispose of in accordance with local regulations. Dispose of waste in accordance with environmental legislation. |
| <b>Contaminated packaging</b>              | Do not reuse empty containers.                                                                                  |

**14. Transport information**

|             |               |
|-------------|---------------|
| <u>DOT</u>  | Not regulated |
| <u>TDG</u>  | Not regulated |
| <u>MEX</u>  | Not regulated |
| <u>IATA</u> | Not regulated |
| <u>IMDG</u> | Not regulated |

## 15. Regulatory information

International Inventories Contact supplier for inventory compliance status

### US Federal Regulations

#### SARA 313

Section 313 of Title III of the Superfund Amendments and Reauthorization Act of 1986 (SARA). This product does not contain any chemicals which are subject to the reporting requirements of the Act and Title 40 of the Code of Federal Regulations, Part 372.

#### SARA 311/312 Hazard Categories

Should this product meet EPCRA 311/312 Tier reporting criteria at 40 CFR 370, refer to Section 2 of this SDS for appropriate classifications.

#### CWA (Clean Water Act)

This product does not contain any substances regulated as pollutants pursuant to the Clean Water Act (40 CFR 122.21 and 40 CFR 122.42).

#### CERCLA

This material, as supplied, does not contain any substances regulated as hazardous substances under the Comprehensive Environmental Response Compensation and Liability Act (CERCLA) (40 CFR 302) or the Superfund Amendments and Reauthorization Act (SARA) (40 CFR 355). There may be specific reporting requirements at the local, regional, or state level pertaining to releases of this material.

### US State Regulations

#### California Proposition 65

This product does not contain any Proposition 65 chemicals.

#### U.S. State Right-to-Know Regulations

This product does not contain any substances regulated under applicable state right-to-know regulations

### U.S. EPA Label Information

**EPA Pesticide Registration Number** Not applicable

## 16. Other information

|             |                  |                |                    |                       |
|-------------|------------------|----------------|--------------------|-----------------------|
| <u>NFPA</u> | Health hazards 0 | Flammability 0 | Instability 0      | Special hazards -     |
| <u>HMIS</u> | Health hazards 0 | Flammability 0 | Physical hazards 0 | Personal protection X |

#### Key or legend to abbreviations and acronyms used in the safety data sheet

##### Legend Section 8: EXPOSURE CONTROLS/PERSONAL PROTECTION

|         |                             |      |                                  |
|---------|-----------------------------|------|----------------------------------|
| TWA     | TWA (time-weighted average) | STEL | STEL (Short Term Exposure Limit) |
| Ceiling | Maximum limit value         | *    | Skin designation                 |



**Key literature references and sources for data used to compile the SDS**

Agency for Toxic Substances and Disease Registry (ATSDR)  
U.S. Environmental Protection Agency ChemView Database  
European Food Safety Authority (EFSA)  
EPA (Environmental Protection Agency)  
Acute Exposure Guideline Level(s) (AEGl(s))  
U.S. Environmental Protection Agency Federal Insecticide, Fungicide, and Rodenticide Act  
U.S. Environmental Protection Agency High Production Volume Chemicals  
Food Research Journal  
Hazardous Substance Database  
International Uniform Chemical Information Database (IUCLID)  
National Institute of Technology and Evaluation (NITE)  
Australia National Industrial Chemicals Notification and Assessment Scheme (NICNAS)  
NIOSH (National Institute for Occupational Safety and Health)  
National Library of Medicine's ChemID Plus (NLM CIP)  
National Library of Medicine's PubMed database (NLM PUBMED)  
National Toxicology Program (NTP)  
New Zealand's Chemical Classification and Information Database (CCID)  
Organization for Economic Co-operation and Development Environment, Health, and Safety Publications  
Organization for Economic Co-operation and Development High Production Volume Chemicals Program  
Organization for Economic Co-operation and Development Screening Information Data Set  
World Health Organization

**Revision date** 03-Mar-2022

**Revision Note** Significant changes throughout SDS. Review all sections.

**Disclaimer**

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**End of Safety Data Sheet**

# SAFETY DATA SHEET



Revision date 03-Mar-2022

Revision Number 1

## 1. Identification

### Product identifier

**Product Name** R3 - Negative Control (1 ml)

### Other means of identification

**Catalog Number(s)** 7256C

### Recommended use of the chemical and restrictions on use

**Recommended use** Restricted to professional users In vitro diagnostic

### Details of the supplier of the safety data sheet

#### Corporate Headquarters

Bio-Rad Laboratories Inc.  
1000 Alfred Nobel Drive  
Hercules, CA 94547  
USA

#### Manufacturer Address

Bio-Rad  
3 boulevard Raymond Poincaré  
92430 Marnes-la-Coquette  
France  
e-mail: fds-msds.fr@bio-rad.com

#### Legal Entity / Contact Address

Bio-Rad Laboratories  
Clinical Diagnostics  
4000 Alfred Nobel Drive  
Hercules, California 94547  
USA

**Technical Service** 1-800-224-6723  
TechSupport.USSD@bio-rad.com

### Emergency telephone number

**24 Hour Emergency Phone Number** CHEMTREC USA: 1 (800) 424-9300

## 2. Hazard(s) identification

### Classification

|                    |             |
|--------------------|-------------|
| Skin sensitization | Category 1A |
|--------------------|-------------|

### Hazards not otherwise classified (HNOC)

Not applicable

### Label elements

#### Warning

#### Hazard statements

May cause an allergic skin reaction



**Appearance** Liquid

**Physical state** Liquid

**Odor** Odorless

**Precautionary Statements - Prevention**

Avoid breathing dust/fume/gas/mist/vapors/spray  
Contaminated work clothing must not be allowed out of the workplace  
Wear protective gloves

IF ON SKIN: Wash with plenty of water and soap  
If skin irritation or rash occurs: Get medical advice/attention  
Wash contaminated clothing before reuse

**Precautionary Statements - Disposal**

Dispose of contents/container in accordance with local, regional, national, and international regulations as applicable

**Other information**

Harmful to aquatic life with long lasting effects. Harmful to aquatic life. Contains animal source material. (Cattle).  
Contains human source material and / or potentially infectious components

**3. Composition/information on ingredients****Substance**

Not applicable.

**Mixture**

| Chemical name                                                                       | CAS No     | Weight-%     | Trade secret |
|-------------------------------------------------------------------------------------|------------|--------------|--------------|
| 1,2,3-Propanetriol                                                                  | 56-81-5    | 20 - 35      | *            |
| 5-Chloro-2-methyl-3(2H)-isothiazolone, mixture with<br>2-methyl-3(2H)-isothiazolone | 55965-84-9 | 0.001 - 0.01 | *            |

\*The exact percentage (concentration) of composition has been withheld as a trade secret.

**4. First-aid measures****Description of first aid measures**

|                       |                                                                                                                                                                                       |
|-----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>General advice</b> | Show this safety data sheet to the doctor in attendance.                                                                                                                              |
| <b>Inhalation</b>     | Remove to fresh air.                                                                                                                                                                  |
| <b>Eye contact</b>    | Contains human source material and / or potentially infectious components. Call a physician. Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. |
| <b>Skin contact</b>   | Wash with soap and water. May cause an allergic skin reaction. In the case of skin irritation or allergic reactions see a physician.                                                  |
| <b>Ingestion</b>      | Contains human source material and / or potentially infectious components. Call a physician.                                                                                          |

**Most important symptoms and effects, both acute and delayed**

**Symptoms** Itching. Rashes. Hives.

**Indication of any immediate medical attention and special treatment needed**

**Note to physicians** May cause sensitization in susceptible persons. Treat symptomatically. Contains human source material and / or potentially infectious components.

**5. Fire-fighting measures**

|                                                                       |                                                                                                                                    |
|-----------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------|
| <b>Suitable Extinguishing Media</b>                                   | Use extinguishing measures that are appropriate to local circumstances and the surrounding environment.                            |
| <b>Specific hazards arising from the chemical</b>                     | Product is or contains a sensitizer. May cause sensitization by skin contact.                                                      |
| <b>Explosion data</b>                                                 |                                                                                                                                    |
| <b>Sensitivity to mechanical impact</b>                               | None.                                                                                                                              |
| <b>Sensitivity to static discharge</b>                                | None.                                                                                                                              |
| <b>Special protective equipment and precautions for fire-fighters</b> | Firefighters should wear self-contained breathing apparatus and full firefighting turnout gear. Use personal protection equipment. |

## 6. Accidental release measures

### Personal precautions, protective equipment and emergency procedures

|                             |                                                                                                                                                                                                          |
|-----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Personal precautions</b> | Avoid contact with skin, eyes or clothing. Ensure adequate ventilation. Use personal protective equipment as required. Evacuate personnel to safe areas. Keep people away from and upwind of spill/leak. |
|-----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

### Methods and material for containment and cleaning up

|                                |                                                                       |
|--------------------------------|-----------------------------------------------------------------------|
| <b>Methods for containment</b> | Do not allow into any sewer, on the ground or into any body of water. |
| <b>Methods for cleaning up</b> | Use: Disinfectant. Clean contaminated surface thoroughly.             |

## 7. Handling and storage

### Precautions for safe handling

|                                |                                                                                                                                                                                                                                                                                                                                    |
|--------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Advice on safe handling</b> | Handle in accordance with good industrial hygiene and safety practice. Avoid contact with skin, eyes or clothing. Ensure adequate ventilation. In case of insufficient ventilation, wear suitable respiratory equipment. Do not eat, drink or smoke when using this product. Take off contaminated clothing and wash before reuse. |
|--------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

### Conditions for safe storage, including any incompatibilities

|                           |                                                                                                                             |
|---------------------------|-----------------------------------------------------------------------------------------------------------------------------|
| <b>Storage Conditions</b> | Keep containers tightly closed in a dry, cool and well-ventilated place. Store according to product and label instructions. |
|---------------------------|-----------------------------------------------------------------------------------------------------------------------------|

## 8. Exposure controls/personal protection

### Control parameters

|                        |                                                                                                                                                                                                                                                                                                                                                                                                            |
|------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Exposure Limits</b> | The following ingredients are the only ingredients of the product above the cut-off level (or level that contributes to the hazard classification of the mixture) which have an exposure limit applicable in the region for which this safety data sheet is intended or other recommended limit. At this time, the other relevant constituents have no known exposure limits from the sources listed here. |
|------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

| Chemical name      | ACGIH TLV | OSHA PEL                              | NIOSH |
|--------------------|-----------|---------------------------------------|-------|
| 1,2,3-Propanetriol | -         | TWA: 15 mg/m <sup>3</sup> mist, total | -     |

|         |  |                                                                                                                                                                                                  |  |
|---------|--|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
| 56-81-5 |  | particulate<br>TWA: 5 mg/m <sup>3</sup> mist, respirable fraction<br>(vacated) TWA: 10 mg/m <sup>3</sup> mist, total particulate<br>(vacated) TWA: 5 mg/m <sup>3</sup> mist, respirable fraction |  |
|---------|--|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|

**Appropriate engineering controls**

**Engineering controls**                      Showers  
                                                          Eyewash stations  
                                                          Ventilation systems.

**Individual protection measures, such as personal protective equipment**

**Eye/face protection**                      Wear safety glasses with side shields (or goggles).

**Hand protection**                              Wear suitable gloves.

**Skin and body protection**                      Wear suitable protective clothing.

**Respiratory protection**                      No protective equipment is needed under normal use conditions. If exposure limits are exceeded or irritation is experienced, ventilation and evacuation may be required.

**General hygiene considerations**                      Follow universal and standard precautions for handling potentially infectious materials.

## 9. Physical and chemical properties

**Information on basic physical and chemical properties**

**Physical state**                                      Liquid  
**Appearance**                                      Liquid  
**Color**                                                  orange  
**Odor**                                                  Odorless  
**Odor threshold**                                      No information available

| <b><u>Property</u></b>                        | <b><u>Values</u></b> | <b><u>Remarks • Method</u></b> |
|-----------------------------------------------|----------------------|--------------------------------|
| <b>pH</b>                                     |                      |                                |
| <b>Melting point / freezing point</b>         | No data available    | None known                     |
| <b>Boiling point / boiling range</b>          | No data available    | None known                     |
| <b>Flash point</b>                            | No data available    | None known                     |
| <b>Evaporation rate</b>                       | No data available    | None known                     |
| <b>Flammability (solid, gas)</b>              | No data available    | None known                     |
| <b>Flammability Limit in Air</b>              |                      | None known                     |
| <b>Upper flammability or explosive limits</b> | No data available    |                                |
| <b>Lower flammability or explosive limits</b> | No data available    |                                |
| <b>Vapor pressure</b>                         | No data available    | None known                     |
| <b>Vapor density</b>                          | No data available    | None known                     |
| <b>Relative density</b>                       | No data available    | None known                     |
| <b>Water solubility</b>                       | Miscible in water    |                                |
| <b>Solubility(ies)</b>                        | No data available    | None known                     |
| <b>Partition coefficient</b>                  | No data available    | None known                     |
| <b>Autoignition temperature</b>               | 392.78               | None known                     |
| <b>Decomposition temperature</b>              |                      | None known                     |
| <b>Kinematic viscosity</b>                    | No data available    | None known                     |
| <b>Dynamic viscosity</b>                      | No data available    | None known                     |

**Other information**

**Explosive properties**                              No information available

|                             |                          |
|-----------------------------|--------------------------|
| <b>Oxidizing properties</b> | No information available |
| <b>Softening point</b>      | No information available |
| <b>Molecular weight</b>     | No information available |
| <b>VOC Content (%)</b>      | No information available |
| <b>Liquid Density</b>       | No information available |
| <b>Bulk density</b>         | No information available |

## 10. Stability and reactivity

|                                           |                                           |
|-------------------------------------------|-------------------------------------------|
| <b>Reactivity</b>                         | No information available.                 |
| <b>Chemical stability</b>                 | Stable under normal conditions.           |
| <b>Possibility of hazardous reactions</b> | None under normal processing.             |
| <b>Conditions to avoid</b>                | None known based on information supplied. |
| <b>Incompatible materials</b>             | None known based on information supplied. |
| <b>Hazardous decomposition products</b>   | None known based on information supplied. |

## 11. Toxicological information

### Information on likely routes of exposure

#### Product Information

|                     |                                                                                                                                                                                                                             |
|---------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Inhalation</b>   | Specific test data for the substance or mixture is not available.                                                                                                                                                           |
| <b>Eye contact</b>  | Specific test data for the substance or mixture is not available.                                                                                                                                                           |
| <b>Skin contact</b> | May cause sensitization by skin contact. Specific test data for the substance or mixture is not available. Repeated or prolonged skin contact may cause allergic reactions with susceptible persons. (based on components). |
| <b>Ingestion</b>    | Specific test data for the substance or mixture is not available.                                                                                                                                                           |

### Symptoms related to the physical, chemical and toxicological characteristics

|                 |                         |
|-----------------|-------------------------|
| <b>Symptoms</b> | Itching. Rashes. Hives. |
|-----------------|-------------------------|

### Acute toxicity

#### Numerical measures of toxicity

#### Component Information

| Chemical name                                                                                     | Oral LD50             | Dermal LD50              | Inhalation LC50         |
|---------------------------------------------------------------------------------------------------|-----------------------|--------------------------|-------------------------|
| 1,2,3-Propanetriol<br>56-81-5                                                                     | = 12600 mg/kg ( Rat ) | > 10 g/kg ( Rabbit )     | > 2.75 mg/L ( Rat ) 4 h |
| 5-Chloro-2-methyl-3(2H)-isothiazolone, mixture with<br>2-methyl-3(2H)-isothiazolone<br>55965-84-9 | = 53 mg/kg ( Rat )    | = 87.12 mg/kg ( Rabbit ) | -                       |

### Delayed and immediate effects as well as chronic effects from short and long-term exposure

|                                          |                           |
|------------------------------------------|---------------------------|
| <b>Skin corrosion/irritation</b>         | No information available. |
| <b>Serious eye damage/eye irritation</b> | No information available. |

|                                   |                                          |
|-----------------------------------|------------------------------------------|
| Respiratory or skin sensitization | May cause sensitization by skin contact. |
| Germ cell mutagenicity            | No information available.                |
| Carcinogenicity                   | No information available.                |
| Reproductive toxicity             | No information available.                |
| STOT - single exposure            | No information available.                |
| STOT - repeated exposure          | No information available.                |
| Target organ effects              | Kidney, Respiratory system, Eyes, Skin.  |
| Aspiration hazard                 | No information available.                |
| Other adverse effects             | No information available.                |
| Interactive effects               | No information available.                |

## 12. Ecological information

**Ecotoxicity** Harmful to aquatic life with long lasting effects.

| Chemical name                 | Algae/aquatic plants | Fish                                            | Toxicity to microorganisms | Crustacea |
|-------------------------------|----------------------|-------------------------------------------------|----------------------------|-----------|
| 1,2,3-Propanetriol<br>56-81-5 | -                    | LC50: 51 - 57mL/L (96h,<br>Oncorhynchus mykiss) | -                          | -         |

**Persistence and degradability** No information available.

### Bioaccumulation

#### Component Information

| Chemical name                 | Partition coefficient |
|-------------------------------|-----------------------|
| 1,2,3-Propanetriol<br>56-81-5 | -1.76                 |

**Other adverse effects** No information available.

## 13. Disposal considerations

### Waste treatment methods

|                                     |                                                                                                                 |
|-------------------------------------|-----------------------------------------------------------------------------------------------------------------|
| Waste from residues/unused products | Dispose of in accordance with local regulations. Dispose of waste in accordance with environmental legislation. |
| Contaminated packaging              | Do not reuse empty containers.                                                                                  |

**14. Transport information**

|             |               |
|-------------|---------------|
| <u>DOT</u>  | Not regulated |
| <u>TDG</u>  | Not regulated |
| <u>MEX</u>  | Not regulated |
| <u>IATA</u> | Not regulated |
| <u>IMDG</u> | Not regulated |

**15. Regulatory information**

International Inventories Contact supplier for inventory compliance status

**US Federal Regulations****SARA 313**

Section 313 of Title III of the Superfund Amendments and Reauthorization Act of 1986 (SARA). This product does not contain any chemicals which are subject to the reporting requirements of the Act and Title 40 of the Code of Federal Regulations, Part 372.

**SARA 311/312 Hazard Categories**

Should this product meet EPCRA 311/312 Tier reporting criteria at 40 CFR 370, refer to Section 2 of this SDS for appropriate classifications.

**CWA (Clean Water Act)**

This product does not contain any substances regulated as pollutants pursuant to the Clean Water Act (40 CFR 122.21 and 40 CFR 122.42).

**CERCLA**

This material, as supplied, does not contain any substances regulated as hazardous substances under the Comprehensive Environmental Response Compensation and Liability Act (CERCLA) (40 CFR 302) or the Superfund Amendments and Reauthorization Act (SARA) (40 CFR 355). There may be specific reporting requirements at the local, regional, or state level pertaining to releases of this material.

**US State Regulations****California Proposition 65**

This product does not contain any Proposition 65 chemicals.

**U.S. State Right-to-Know Regulations**

| Chemical name                 | New Jersey | Massachusetts | Pennsylvania |
|-------------------------------|------------|---------------|--------------|
| Water<br>7732-18-5            | -          | -             | X            |
| 1,2,3-Propanetriol<br>56-81-5 | X          | X             | X            |

**U.S. EPA Label Information**

**EPA Pesticide Registration Number** Not applicable



## 16. Other information

|             |                         |                       |                           |                              |
|-------------|-------------------------|-----------------------|---------------------------|------------------------------|
| <b>NFPA</b> | <b>Health hazards</b> 2 | <b>Flammability</b> 1 | <b>Instability</b> 0      | <b>Special hazards</b> -     |
| <b>HMIS</b> | <b>Health hazards</b> 2 | <b>Flammability</b> 1 | <b>Physical hazards</b> 0 | <b>Personal protection</b> X |

### Key or legend to abbreviations and acronyms used in the safety data sheet

#### **Legend Section 8: EXPOSURE CONTROLS/PERSONAL PROTECTION**

|         |                             |      |                                  |
|---------|-----------------------------|------|----------------------------------|
| TWA     | TWA (time-weighted average) | STEL | STEL (Short Term Exposure Limit) |
| Ceiling | Maximum limit value         | *    | Skin designation                 |

#### **Key literature references and sources for data used to compile the SDS**

Agency for Toxic Substances and Disease Registry (ATSDR)  
 U.S. Environmental Protection Agency ChemView Database  
 European Food Safety Authority (EFSA)  
 EPA (Environmental Protection Agency)  
 Acute Exposure Guideline Level(s) (AEGl(s))  
 U.S. Environmental Protection Agency Federal Insecticide, Fungicide, and Rodenticide Act  
 U.S. Environmental Protection Agency High Production Volume Chemicals  
 Food Research Journal  
 Hazardous Substance Database  
 International Uniform Chemical Information Database (IUCLID)  
 National Institute of Technology and Evaluation (NITE)  
 Australia National Industrial Chemicals Notification and Assessment Scheme (NICNAS)  
 NIOSH (National Institute for Occupational Safety and Health)  
 National Library of Medicine's ChemID Plus (NLM CIP)  
 National Library of Medicine's PubMed database (NLM PUBMED)  
 National Toxicology Program (NTP)  
 New Zealand's Chemical Classification and Information Database (CCID)  
 Organization for Economic Co-operation and Development Environment, Health, and Safety Publications  
 Organization for Economic Co-operation and Development High Production Volume Chemicals Program  
 Organization for Economic Co-operation and Development Screening Information Data Set  
 World Health Organization

**Revision date** 03-Mar-2022

**Revision Note** Significant changes throughout SDS. Review all sections.

#### **Disclaimer**

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text.

**End of Safety Data Sheet**

# SAFETY DATA SHEET



Revision date 03-Mar-2022

Revision Number 1

## 1. Identification

### Product identifier

**Product Name** R4 - Ab Positive Control (1,5 ml)

### Other means of identification

**Catalog Number(s)** 7256E

### Recommended use of the chemical and restrictions on use

**Recommended use** Restricted to professional users In vitro diagnostic

### Details of the supplier of the safety data sheet

#### Corporate Headquarters

Bio-Rad Laboratories Inc.  
1000 Alfred Nobel Drive  
Hercules, CA 94547  
USA

#### Manufacturer Address

Bio-Rad  
3 boulevard Raymond Poincaré  
92430 Marnes-la-Coquette  
France  
e-mail: fds-msds.fr@bio-rad.com

#### Legal Entity / Contact Address

Bio-Rad Laboratories  
Clinical Diagnostics  
4000 Alfred Nobel Drive  
Hercules, California 94547  
USA

**Technical Service** 1-800-224-6723  
TechSupport.USSD@bio-rad.com

### Emergency telephone number

**24 Hour Emergency Phone Number** CHEMTREC USA: 1 (800) 424-9300

## 2. Hazard(s) identification

### Classification

|                    |             |
|--------------------|-------------|
| Skin sensitization | Category 1A |
|--------------------|-------------|

### Hazards not otherwise classified (HNOC)

Not applicable

### Label elements

#### Warning

#### Hazard statements

May cause an allergic skin reaction



**Appearance** Liquid

**Physical state** Liquid

**Odor** Odorless

**Precautionary Statements - Prevention**

Avoid breathing dust/fume/gas/mist/vapors/spray  
Contaminated work clothing must not be allowed out of the workplace  
Wear protective gloves

IF ON SKIN: Wash with plenty of water and soap  
If skin irritation or rash occurs: Get medical advice/attention  
Wash contaminated clothing before reuse

**Precautionary Statements - Disposal**

Dispose of contents/container in accordance with local, regional, national, and international regulations as applicable

**Other information**

Harmful to aquatic life with long lasting effects. Harmful to aquatic life. Contains animal source material. (Cattle).

**3. Composition/information on ingredients****Substance**

Not applicable.

**Mixture**

| Chemical name                                                                    | CAS No     | Weight-%     | Trade secret |
|----------------------------------------------------------------------------------|------------|--------------|--------------|
| 1,2,3-Propanetriol                                                               | 56-81-5    | 20 - 35      | *            |
| 5-Chloro-2-methyl-3(2H)-isothiazolone, mixture with 2-methyl-3(2H)-isothiazolone | 55965-84-9 | 0.001 - 0.01 | *            |

\*The exact percentage (concentration) of composition has been withheld as a trade secret.

**4. First-aid measures****Description of first aid measures**

|                       |                                                                                                                                      |
|-----------------------|--------------------------------------------------------------------------------------------------------------------------------------|
| <b>General advice</b> | Show this safety data sheet to the doctor in attendance.                                                                             |
| <b>Inhalation</b>     | Remove to fresh air.                                                                                                                 |
| <b>Eye contact</b>    | Rinse thoroughly with plenty of water for at least 15 minutes, lifting lower and upper eyelids. Consult a physician.                 |
| <b>Skin contact</b>   | Wash with soap and water. May cause an allergic skin reaction. In the case of skin irritation or allergic reactions see a physician. |
| <b>Ingestion</b>      | Rinse mouth.                                                                                                                         |

**Most important symptoms and effects, both acute and delayed**

|                 |                         |
|-----------------|-------------------------|
| <b>Symptoms</b> | Itching. Rashes. Hives. |
|-----------------|-------------------------|

**Indication of any immediate medical attention and special treatment needed**

|                           |                                                                        |
|---------------------------|------------------------------------------------------------------------|
| <b>Note to physicians</b> | May cause sensitization in susceptible persons. Treat symptomatically. |
|---------------------------|------------------------------------------------------------------------|

**5. Fire-fighting measures**

|                                     |                                                                                                         |
|-------------------------------------|---------------------------------------------------------------------------------------------------------|
| <b>Suitable Extinguishing Media</b> | Use extinguishing measures that are appropriate to local circumstances and the surrounding environment. |
|-------------------------------------|---------------------------------------------------------------------------------------------------------|

**Specific hazards arising from the chemical** Product is or contains a sensitizer. May cause sensitization by skin contact.

**Explosion data**

**Sensitivity to mechanical impact** None.

**Sensitivity to static discharge** None.

**Special protective equipment and precautions for fire-fighters** Firefighters should wear self-contained breathing apparatus and full firefighting turnout gear. Use personal protection equipment.

## 6. Accidental release measures

### Personal precautions, protective equipment and emergency procedures

**Personal precautions** Avoid contact with skin, eyes or clothing. Ensure adequate ventilation. Use personal protective equipment as required. Evacuate personnel to safe areas. Keep people away from and upwind of spill/leak.

### Methods and material for containment and cleaning up

**Methods for containment** Prevent further leakage or spillage if safe to do so.

**Methods for cleaning up** Pick up and transfer to properly labeled containers.

## 7. Handling and storage

### Precautions for safe handling

**Advice on safe handling** Handle in accordance with good industrial hygiene and safety practice. Avoid contact with skin, eyes or clothing. Ensure adequate ventilation. In case of insufficient ventilation, wear suitable respiratory equipment. Do not eat, drink or smoke when using this product. Take off contaminated clothing and wash before reuse.

### Conditions for safe storage, including any incompatibilities

**Storage Conditions** Keep containers tightly closed in a dry, cool and well-ventilated place. Store according to product and label instructions.

## 8. Exposure controls/personal protection

### Control parameters

**Exposure Limits** The following ingredients are the only ingredients of the product above the cut-off level (or level that contributes to the hazard classification of the mixture) which have an exposure limit applicable in the region for which this safety data sheet is intended or other recommended limit. At this time, the other relevant constituents have no known exposure limits from the sources listed here.

| Chemical name                 | ACGIH TLV | OSHA PEL                                                                                                                                       | NIOSH |
|-------------------------------|-----------|------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| 1,2,3-Propanetriol<br>56-81-5 | -         | TWA: 15 mg/m <sup>3</sup> mist, total particulate<br>TWA: 5 mg/m <sup>3</sup> mist, respirable fraction<br>(vacated) TWA: 10 mg/m <sup>3</sup> | -     |



|                |                          |
|----------------|--------------------------|
| Liquid Density | No information available |
| Bulk density   | No information available |

## 10. Stability and reactivity

|                                    |                                           |
|------------------------------------|-------------------------------------------|
| Reactivity                         | No information available.                 |
| Chemical stability                 | Stable under normal conditions.           |
| Possibility of hazardous reactions | None under normal processing.             |
| Conditions to avoid                | None known based on information supplied. |
| Incompatible materials             | None known based on information supplied. |
| Hazardous decomposition products   | None known based on information supplied. |

## 11. Toxicological information

### Information on likely routes of exposure

#### Product Information

|              |                                                                                                                                                                                                                             |
|--------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Inhalation   | Specific test data for the substance or mixture is not available.                                                                                                                                                           |
| Eye contact  | Specific test data for the substance or mixture is not available.                                                                                                                                                           |
| Skin contact | May cause sensitization by skin contact. Specific test data for the substance or mixture is not available. Repeated or prolonged skin contact may cause allergic reactions with susceptible persons. (based on components). |
| Ingestion    | Specific test data for the substance or mixture is not available.                                                                                                                                                           |

### Symptoms related to the physical, chemical and toxicological characteristics

|          |                         |
|----------|-------------------------|
| Symptoms | Itching. Rashes. Hives. |
|----------|-------------------------|

### Acute toxicity

#### Numerical measures of toxicity

#### Component Information

| Chemical name                                                                                     | Oral LD50             | Dermal LD50              | Inhalation LC50         |
|---------------------------------------------------------------------------------------------------|-----------------------|--------------------------|-------------------------|
| 1,2,3-Propanetriol<br>56-81-5                                                                     | = 12600 mg/kg ( Rat ) | > 10 g/kg ( Rabbit )     | > 2.75 mg/L ( Rat ) 4 h |
| 5-Chloro-2-methyl-3(2H)-isothiazolone, mixture with<br>2-methyl-3(2H)-isothiazolone<br>55965-84-9 | = 53 mg/kg ( Rat )    | = 87.12 mg/kg ( Rabbit ) | -                       |

### Delayed and immediate effects as well as chronic effects from short and long-term exposure

|                                   |                                          |
|-----------------------------------|------------------------------------------|
| Skin corrosion/irritation         | No information available.                |
| Serious eye damage/eye irritation | No information available.                |
| Respiratory or skin sensitization | May cause sensitization by skin contact. |

|                          |                                         |
|--------------------------|-----------------------------------------|
| Germ cell mutagenicity   | No information available.               |
| Carcinogenicity          | No information available.               |
| Reproductive toxicity    | No information available.               |
| STOT - single exposure   | No information available.               |
| STOT - repeated exposure | No information available.               |
| Target organ effects     | Kidney, Respiratory system, Eyes, Skin. |
| Aspiration hazard        | No information available.               |
| Other adverse effects    | No information available.               |
| Interactive effects      | No information available.               |

## 12. Ecological information

**Ecotoxicity** Harmful to aquatic life with long lasting effects.

| Chemical name                 | Algae/aquatic plants | Fish                                            | Toxicity to microorganisms | Crustacea |
|-------------------------------|----------------------|-------------------------------------------------|----------------------------|-----------|
| 1,2,3-Propanetriol<br>56-81-5 | -                    | LC50: 51 - 57mL/L (96h,<br>Oncorhynchus mykiss) | -                          | -         |

**Persistence and degradability** No information available.

### Bioaccumulation

#### Component Information

| Chemical name                 | Partition coefficient |
|-------------------------------|-----------------------|
| 1,2,3-Propanetriol<br>56-81-5 | -1.76                 |

**Other adverse effects** No information available.

## 13. Disposal considerations

### Waste treatment methods

**Waste from residues/unused products** Dispose of in accordance with local regulations. Dispose of waste in accordance with environmental legislation.

**Contaminated packaging** Do not reuse empty containers.

## 14. Transport information

|             |               |
|-------------|---------------|
| <u>DOT</u>  | Not regulated |
| <u>TDG</u>  | Not regulated |
| <u>MEX</u>  | Not regulated |
| <u>IATA</u> | Not regulated |
| <u>IMDG</u> | Not regulated |

## 15. Regulatory information

International Inventories Contact supplier for inventory compliance status

### US Federal Regulations

#### **SARA 313**

Section 313 of Title III of the Superfund Amendments and Reauthorization Act of 1986 (SARA). This product does not contain any chemicals which are subject to the reporting requirements of the Act and Title 40 of the Code of Federal Regulations, Part 372.

#### **SARA 311/312 Hazard Categories**

Should this product meet EPCRA 311/312 Tier reporting criteria at 40 CFR 370, refer to Section 2 of this SDS for appropriate classifications.

#### **CWA (Clean Water Act)**

This product does not contain any substances regulated as pollutants pursuant to the Clean Water Act (40 CFR 122.21 and 40 CFR 122.42).

#### **CERCLA**

This material, as supplied, does not contain any substances regulated as hazardous substances under the Comprehensive Environmental Response Compensation and Liability Act (CERCLA) (40 CFR 302) or the Superfund Amendments and Reauthorization Act (SARA) (40 CFR 355). There may be specific reporting requirements at the local, regional, or state level pertaining to releases of this material.

### US State Regulations

#### **California Proposition 65**

This product does not contain any Proposition 65 chemicals.

### U.S. State Right-to-Know Regulations

| Chemical name                 | New Jersey | Massachusetts | Pennsylvania |
|-------------------------------|------------|---------------|--------------|
| Water<br>7732-18-5            | -          | -             | X            |
| 1,2,3-Propanetriol<br>56-81-5 | X          | X             | X            |

### U.S. EPA Label Information

EPA Pesticide Registration Number Not applicable

## 16. Other information

|             |                  |                |                    |                       |
|-------------|------------------|----------------|--------------------|-----------------------|
| <u>NFPA</u> | Health hazards 2 | Flammability 1 | Instability 0      | Special hazards -     |
| <u>HMIS</u> | Health hazards 2 | Flammability 1 | Physical hazards 0 | Personal protection X |



**Key or legend to abbreviations and acronyms used in the safety data sheet****Legend Section 8: EXPOSURE CONTROLS/PERSONAL PROTECTION**

|         |                             |      |                                  |
|---------|-----------------------------|------|----------------------------------|
| TWA     | TWA (time-weighted average) | STEL | STEL (Short Term Exposure Limit) |
| Ceiling | Maximum limit value         | *    | Skin designation                 |

**Key literature references and sources for data used to compile the SDS**

Agency for Toxic Substances and Disease Registry (ATSDR)  
U.S. Environmental Protection Agency ChemView Database  
European Food Safety Authority (EFSA)  
EPA (Environmental Protection Agency)  
Acute Exposure Guideline Level(s) (AEGl(s))  
U.S. Environmental Protection Agency Federal Insecticide, Fungicide, and Rodenticide Act  
U.S. Environmental Protection Agency High Production Volume Chemicals  
Food Research Journal  
Hazardous Substance Database  
International Uniform Chemical Information Database (IUCLID)  
National Institute of Technology and Evaluation (NITE)  
Australia National Industrial Chemicals Notification and Assessment Scheme (NICNAS)  
NIOSH (National Institute for Occupational Safety and Health)  
National Library of Medicine's ChemID Plus (NLM CIP)  
National Library of Medicine's PubMed database (NLM PUBMED)  
National Toxicology Program (NTP)  
New Zealand's Chemical Classification and Information Database (CCID)  
Organization for Economic Co-operation and Development Environment, Health, and Safety Publications  
Organization for Economic Co-operation and Development High Production Volume Chemicals Program  
Organization for Economic Co-operation and Development Screening Information Data Set  
World Health Organization

**Revision date** 03-Mar-2022

**Revision Note** Significant changes throughout SDS. Review all sections.

**Disclaimer**

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text.

**End of Safety Data Sheet**

# SAFETY DATA SHEET



Revision date 03-Mar-2022

Revision Number 1

## 1. Identification

### Product identifier

**Product Name** R5a - Ag Positive Control (q.s. ad 1 ml)

### Other means of identification

**Catalog Number(s)** 7256G

### Recommended use of the chemical and restrictions on use

**Recommended use** Restricted to professional users In vitro diagnostic

### Details of the supplier of the safety data sheet

#### Corporate Headquarters

Bio-Rad Laboratories Inc.  
1000 Alfred Nobel Drive  
Hercules, CA 94547  
USA

#### Manufacturer Address

Bio-Rad  
3 boulevard Raymond Poincaré  
92430 Marnes-la-Coquette  
France  
e-mail: fds-msds.fr@bio-rad.com

#### Legal Entity / Contact Address

Bio-Rad Laboratories  
Clinical Diagnostics  
4000 Alfred Nobel Drive  
Hercules, California 94547  
USA

**Technical Service** 1-800-224-6723  
TechSupport.USSD@bio-rad.com

### Emergency telephone number

**24 Hour Emergency Phone Number** CHEMTREC USA: 1 (800) 424-9300

## 2. Hazard(s) identification

### Classification

This chemical is not considered hazardous by the 2012 OSHA Hazard Communication Standard (29 CFR 1910.1200)

### Hazards not otherwise classified (HNOC)

Not applicable

### Label elements

#### Hazard statements

This chemical is not considered hazardous by the 2012 OSHA Hazard Communication Standard (29 CFR 1910.1200)

The product contains no substances which at their given concentration, are considered to be hazardous to health.

**Appearance** Powder

**Physical state** Solid

**Odor** Odorless

### Other information

Contains animal source material. (Cattle).

### 3. Composition/information on ingredients

**Substance**

Not applicable.

**Mixture**

The product contains no substances which at their given concentration, are considered to be hazardous to health

\*The exact percentage (concentration) of composition has been withheld as a trade secret.

### 4. First-aid measures

**Description of first aid measures**

|              |                                                                                                                         |
|--------------|-------------------------------------------------------------------------------------------------------------------------|
| Inhalation   | Remove to fresh air.                                                                                                    |
| Eye contact  | Rinse thoroughly with plenty of water for at least 15 minutes, lifting lower and upper eyelids.<br>Consult a physician. |
| Skin contact | Wash skin with soap and water.                                                                                          |
| Ingestion    | Rinse mouth.                                                                                                            |

**Most important symptoms and effects, both acute and delayed**

Symptoms No information available.

**Indication of any immediate medical attention and special treatment needed**

Note to physicians Treat symptomatically.

### 5. Fire-fighting measures

|                                                                |                                                                                                                                    |
|----------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------|
| Suitable Extinguishing Media                                   | Use extinguishing measures that are appropriate to local circumstances and the surrounding environment.                            |
| Specific hazards arising from the chemical                     | No information available.                                                                                                          |
| Explosion data                                                 |                                                                                                                                    |
| Sensitivity to mechanical impact                               | None.                                                                                                                              |
| Sensitivity to static discharge                                | None.                                                                                                                              |
| Special protective equipment and precautions for fire-fighters | Firefighters should wear self-contained breathing apparatus and full firefighting turnout gear. Use personal protection equipment. |

### 6. Accidental release measures

**Personal precautions, protective equipment and emergency procedures**

Personal precautions Ensure adequate ventilation.

**Methods and material for containment and cleaning up**

|                         |                                                       |
|-------------------------|-------------------------------------------------------|
| Methods for containment | Prevent further leakage or spillage if safe to do so. |
| Methods for cleaning up | Pick up and transfer to properly labeled containers.  |

**7. Handling and storage****Precautions for safe handling**

|                         |                                                                        |
|-------------------------|------------------------------------------------------------------------|
| Advice on safe handling | Handle in accordance with good industrial hygiene and safety practice. |
|-------------------------|------------------------------------------------------------------------|

**Conditions for safe storage, including any incompatibilities**

|                    |                                                    |
|--------------------|----------------------------------------------------|
| Storage Conditions | Store according to product and label instructions. |
|--------------------|----------------------------------------------------|

**8. Exposure controls/personal protection****Control parameters**

|                 |                                                                                                                                                             |
|-----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Exposure Limits | This product, as supplied, does not contain any hazardous materials with occupational exposure limits established by the region specific regulatory bodies. |
|-----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|

**Appropriate engineering controls**

|                      |                                                     |
|----------------------|-----------------------------------------------------|
| Engineering controls | Showers<br>Eyewash stations<br>Ventilation systems. |
|----------------------|-----------------------------------------------------|

**Individual protection measures, such as personal protective equipment**

|                                |                                                                                                                                                                          |
|--------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Eye/face protection            | No special protective equipment required.                                                                                                                                |
| Skin and body protection       | No special protective equipment required.                                                                                                                                |
| Respiratory protection         | No protective equipment is needed under normal use conditions. If exposure limits are exceeded or irritation is experienced, ventilation and evacuation may be required. |
| General hygiene considerations | Handle in accordance with good industrial hygiene and safety practice.                                                                                                   |

**9. Physical and chemical properties****Information on basic physical and chemical properties**

|                |                          |
|----------------|--------------------------|
| Physical state | Solid                    |
| Appearance     | Powder                   |
| Color          | beige                    |
| Odor           | Odorless                 |
| Odor threshold | No information available |

| <u>Property</u>                | <u>Values</u>     | <u>Remarks • Method</u> |
|--------------------------------|-------------------|-------------------------|
| pH                             |                   | None known              |
| Melting point / freezing point | No data available | None known              |
| Boiling point / boiling range  | No data available | None known              |

|                                        |                          |            |
|----------------------------------------|--------------------------|------------|
| Flash point                            | No data available        | None known |
| Evaporation rate                       | No data available        | None known |
| Flammability (solid, gas)              | No data available        | None known |
| Flammability Limit in Air              |                          | None known |
| Upper flammability or explosive limits | No data available        |            |
| Lower flammability or explosive limits | No data available        |            |
| Vapor pressure                         | No data available        | None known |
| Vapor density                          | No data available        | None known |
| Relative density                       | No data available        | None known |
| Water solubility                       | Soluble in water         |            |
| Solubility(ies)                        | No data available        | None known |
| Partition coefficient                  | No data available        | None known |
| Autoignition temperature               | No data available        | None known |
| Decomposition temperature              |                          | None known |
| Kinematic viscosity                    | No data available        | None known |
| Dynamic viscosity                      | No data available        | None known |
| <b>Other information</b>               |                          |            |
| Explosive properties                   | No information available |            |
| Oxidizing properties                   | No information available |            |
| Softening point                        | No information available |            |
| Molecular weight                       | No information available |            |
| VOC Content (%)                        | No information available |            |
| Liquid Density                         | No information available |            |
| Bulk density                           | No information available |            |

## 10. Stability and reactivity

|                                    |                                           |
|------------------------------------|-------------------------------------------|
| Reactivity                         | No information available.                 |
| Chemical stability                 | Stable under normal conditions.           |
| Possibility of hazardous reactions | None under normal processing.             |
| Conditions to avoid                | None known based on information supplied. |
| Incompatible materials             | None known based on information supplied. |
| Hazardous decomposition products   | None known based on information supplied. |

## 11. Toxicological information

### Information on likely routes of exposure

|              |                                                                   |
|--------------|-------------------------------------------------------------------|
| Inhalation   | Specific test data for the substance or mixture is not available. |
| Eye contact  | Specific test data for the substance or mixture is not available. |
| Skin contact | Specific test data for the substance or mixture is not available. |
| Ingestion    | Specific test data for the substance or mixture is not available. |

### Symptoms related to the physical, chemical and toxicological characteristics

|          |                           |
|----------|---------------------------|
| Symptoms | No information available. |
|----------|---------------------------|

### Acute toxicity

Numerical measures of toxicity

**Delayed and immediate effects as well as chronic effects from short and long-term exposure**

|                                          |                           |
|------------------------------------------|---------------------------|
| <b>Skin corrosion/irritation</b>         | No information available. |
| <b>Serious eye damage/eye irritation</b> | No information available. |
| <b>Respiratory or skin sensitization</b> | No information available. |
| <b>Germ cell mutagenicity</b>            | No information available. |
| <b>Carcinogenicity</b>                   | No information available. |
| <b>Reproductive toxicity</b>             | No information available. |
| <b>STOT - single exposure</b>            | No information available. |
| <b>STOT - repeated exposure</b>          | No information available. |
| <b>Aspiration hazard</b>                 | No information available. |
| <b>Other adverse effects</b>             | No information available. |
| <b>Interactive effects</b>               | No information available. |

**12. Ecological information**

|                                      |                                                                           |
|--------------------------------------|---------------------------------------------------------------------------|
| <b>Ecotoxicity</b>                   | The environmental impact of this product has not been fully investigated. |
| <b>Persistence and degradability</b> | No information available.                                                 |
| <b>Bioaccumulation</b>               | There is no data for this product.                                        |
| <b>Other adverse effects</b>         | No information available.                                                 |

**13. Disposal considerations****Waste treatment methods**

|                                            |                                                                                                                 |
|--------------------------------------------|-----------------------------------------------------------------------------------------------------------------|
| <b>Waste from residues/unused products</b> | Dispose of in accordance with local regulations. Dispose of waste in accordance with environmental legislation. |
| <b>Contaminated packaging</b>              | Do not reuse empty containers.                                                                                  |

**14. Transport information**

|             |               |
|-------------|---------------|
| <u>DOT</u>  | Not regulated |
| <u>TDG</u>  | Not regulated |
| <u>MEX</u>  | Not regulated |
| <u>IATA</u> | Not regulated |
| <u>IMDG</u> | Not regulated |

**15. Regulatory information**

International Inventories Contact supplier for inventory compliance status

**US Federal Regulations****SARA 313**

Section 313 of Title III of the Superfund Amendments and Reauthorization Act of 1986 (SARA). This product does not contain any chemicals which are subject to the reporting requirements of the Act and Title 40 of the Code of Federal Regulations, Part 372.

**SARA 311/312 Hazard Categories**

Should this product meet EPCRA 311/312 Tier reporting criteria at 40 CFR 370, refer to Section 2 of this SDS for appropriate classifications.

**CWA (Clean Water Act)**

This product does not contain any substances regulated as pollutants pursuant to the Clean Water Act (40 CFR 122.21 and 40 CFR 122.42).

**CERCLA**

This material, as supplied, does not contain any substances regulated as hazardous substances under the Comprehensive Environmental Response Compensation and Liability Act (CERCLA) (40 CFR 302) or the Superfund Amendments and Reauthorization Act (SARA) (40 CFR 355). There may be specific reporting requirements at the local, regional, or state level pertaining to releases of this material.

**US State Regulations****California Proposition 65**

This product does not contain any Proposition 65 chemicals.

**U.S. State Right-to-Know Regulations**

| Chemical name      | New Jersey | Massachusetts | Pennsylvania |
|--------------------|------------|---------------|--------------|
| Water<br>7732-18-5 | -          | -             | X            |

**U.S. EPA Label Information**

**EPA Pesticide Registration Number** Not applicable

**16. Other information**

|             |                         |                       |                           |                              |
|-------------|-------------------------|-----------------------|---------------------------|------------------------------|
| <b>NFPA</b> | <b>Health hazards</b> 0 | <b>Flammability</b> 0 | <b>Instability</b> 0      | <b>Special hazards</b> -     |
| <b>HMIS</b> | <b>Health hazards</b> 0 | <b>Flammability</b> 0 | <b>Physical hazards</b> 0 | <b>Personal protection</b> X |

**Key or legend to abbreviations and acronyms used in the safety data sheet****Legend Section 8: EXPOSURE CONTROLS/PERSONAL PROTECTION**

|         |                             |      |                                  |
|---------|-----------------------------|------|----------------------------------|
| TWA     | TWA (time-weighted average) | STEL | STEL (Short Term Exposure Limit) |
| Ceiling | Maximum limit value         | *    | Skin designation                 |

**Key literature references and sources for data used to compile the SDS**

Agency for Toxic Substances and Disease Registry (ATSDR)  
 U.S. Environmental Protection Agency ChemView Database  
 European Food Safety Authority (EFSA)  
 EPA (Environmental Protection Agency)  
 Acute Exposure Guideline Level(s) (AEGl(s))  
 U.S. Environmental Protection Agency Federal Insecticide, Fungicide, and Rodenticide Act  
 U.S. Environmental Protection Agency High Production Volume Chemicals  
 Food Research Journal  
 Hazardous Substance Database  
 International Uniform Chemical Information Database (IUCLID)  
 National Institute of Technology and Evaluation (NITE)  
 Australia National Industrial Chemicals Notification and Assessment Scheme (NICNAS)  
 NIOSH (National Institute for Occupational Safety and Health)  
 National Library of Medicine's ChemID Plus (NLM CIP)  
 National Library of Medicine's PubMed database (NLM PUBMED)  
 National Toxicology Program (NTP)  
 New Zealand's Chemical Classification and Information Database (CCID)  
 Organization for Economic Co-operation and Development Environment, Health, and Safety Publications  
 Organization for Economic Co-operation and Development High Production Volume Chemicals Program  
 Organization for Economic Co-operation and Development Screening Information Data Set  
 World Health Organization

**Revision date** 03-Mar-2022

**Revision Note** Significant changes throughout SDS. Review all sections.

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**End of Safety Data Sheet**





# SAFETY DATA SHEET

Revision date 03-Mar-2022

Revision Number 1

## 1. Identification

### Product identifier

**Product Name** R5b - Ag Positive Control Diluent (1 ml)

### Other means of identification

**Catalog Number(s)** 7256H

### Recommended use of the chemical and restrictions on use

**Recommended use** Restricted to professional users  
In vitro diagnostic

**Restrictions on use** No information available

### Details of the supplier of the safety data sheet

#### **Corporate Headquarters**

Bio-Rad Laboratories Inc.  
1000 Alfred Nobel Drive  
Hercules, CA 94547  
USA

#### **Manufacturer Address**

Bio-Rad  
3 boulevard Raymond Poincaré  
92430 Marnes-la-Coquette  
France  
e-mail: fds-msds.fr@bio-rad.com

#### **Legal Entity / Contact Address**

Bio-Rad Eugenia 197 piso 10 -A  
Col. Narvarte Del. Benito Juarez  
CP. 03020, CDMX  
México

#### **Technical Service**

+5255548870 ext. 1031 & 1083  
cts\_cdg\_mexico@bio-rad.com

### Emergency telephone number

**24 Hour Emergency Phone Number** CHEMTREC México: 01-800-681-9531

## 2. Hazard(s) identification

### Classification

|                    |                      |
|--------------------|----------------------|
| Skin sensitization | Category 1A - (H317) |
|--------------------|----------------------|

### Label elements

#### **Warning**

#### **Hazard statements**

H317 - May cause an allergic skin reaction



Exclamation mark

#### **Precautionary Statements - Prevention**

P261 - Avoid breathing dust/fume/gas/mist/vapors/spray

P272 - Contaminated work clothing should not be allowed out of the workplace

P280 - Wear protective gloves/protective clothing/eye protection/face protection

**Skin**

P302 + P352 - IF ON SKIN: Wash with plenty of water and soap

P333 + P313 - If skin irritation or rash occurs: Get medical advice/attention

P362 + P364 - Take off contaminated clothing and wash it before reuse

**Precautionary Statements - Disposal**

P501 - Dispose of contents/container in accordance with local, regional, national, and international regulations as applicable

**Other information**

Harmful to aquatic life with long lasting effects Harmful to aquatic life

**3. Composition/information on ingredients****Substance**

Not applicable.

**Mixture**

The product contains no substances which at their given concentration, are considered to be hazardous to health

**4. First-aid measures****Description of first aid measures**

|                       |                                                                                                                                      |
|-----------------------|--------------------------------------------------------------------------------------------------------------------------------------|
| <b>General advice</b> | Show this safety data sheet to the doctor in attendance.                                                                             |
| <b>Inhalation</b>     | Remove to fresh air.                                                                                                                 |
| <b>Eye contact</b>    | Rinse thoroughly with plenty of water for at least 15 minutes, lifting lower and upper eyelids. Consult a physician.                 |
| <b>Skin contact</b>   | Wash with soap and water. May cause an allergic skin reaction. In the case of skin irritation or allergic reactions see a physician. |
| <b>Ingestion</b>      | Rinse mouth thoroughly with water.                                                                                                   |

**Most important symptoms and effects, both acute and delayed**

|                 |                      |
|-----------------|----------------------|
| <b>Symptoms</b> | Itching Rashes Hives |
|-----------------|----------------------|

**Indication of any immediate medical attention and special treatment needed**

|                           |                                                                        |
|---------------------------|------------------------------------------------------------------------|
| <b>Note to physicians</b> | May cause sensitization in susceptible persons. Treat symptomatically. |
|---------------------------|------------------------------------------------------------------------|

**5. Fire-fighting measures**

|                                                   |                                                                                                         |
|---------------------------------------------------|---------------------------------------------------------------------------------------------------------|
| <b>Suitable Extinguishing Media</b>               | Use extinguishing measures that are appropriate to local circumstances and the surrounding environment. |
| <b>Unsuitable extinguishing media</b>             | No information available.                                                                               |
| <b>Specific hazards arising from the chemical</b> | Product is or contains a sensitizer. May cause sensitization by skin contact.                           |
| <b>Explosion data</b>                             |                                                                                                         |

Sensitivity to mechanical impact None.

Sensitivity to static discharge None.

**Special protective actions for fire-fighters** Firefighters should wear self-contained breathing apparatus and full firefighting turnout gear. Use personal protection equipment.

## 6. Accidental release measures

### Personal precautions, protective equipment and emergency procedures

**Personal precautions** Avoid contact with skin, eyes or clothing. Ensure adequate ventilation. Use personal protective equipment as required. Evacuate personnel to safe areas. Keep people away from and upwind of spill/leak.

### Environmental precautions

**Environmental precautions** See Section 12 for additional Ecological Information.

### Methods and material for containment and cleaning up

**Methods for containment** Prevent further leakage or spillage if safe to do so.

**Methods for cleaning up** Pick up and transfer to properly labeled containers.

**Prevention of secondary hazards** Clean contaminated objects and areas thoroughly observing environmental regulations.

## 7. Handling and storage

### Precautions for safe handling

**Advice on safe handling** Handle in accordance with good industrial hygiene and safety practice. Avoid contact with skin, eyes or clothing. Ensure adequate ventilation. In case of insufficient ventilation, wear suitable respiratory equipment. Do not eat, drink or smoke when using this product. Take off contaminated clothing and wash before reuse.

### Conditions for safe storage, including any incompatibilities

**Storage Conditions** Keep containers tightly closed in a dry, cool and well-ventilated place. Store locked up. Keep out of the reach of children. Store according to product and label instructions.

## 8. Exposure controls/personal protection

### Control parameters

**Exposure Limits** This product, as supplied, does not contain any hazardous materials with occupational exposure limits established by the region specific regulatory bodies.

### Appropriate engineering controls

**Engineering controls** Showers  
Eyewash stations  
Ventilation systems.

### Individual protection measures, such as personal protective equipment

**Eye/face protection** Wear safety glasses with side shields (or goggles).

**Hand protection** Wear suitable gloves.

**Skin and body protection** Wear suitable protective clothing.

**Respiratory protection** No protective equipment is needed under normal use conditions. If exposure limits are exceeded or irritation is experienced, ventilation and evacuation may be required.

General hygiene considerations      Handle in accordance with good industrial hygiene and safety practice.

## 9. Physical and chemical properties

### Information on basic physical and chemical properties

|                |                          |
|----------------|--------------------------|
| Physical state | Liquid                   |
| Appearance     | Liquid                   |
| Color          | blue                     |
| Odor           | Odorless                 |
| Odor threshold | No information available |

| Property                               | Values            | Remarks • Method |
|----------------------------------------|-------------------|------------------|
| pH                                     |                   | None known       |
| Melting point / freezing point         | No data available | None known       |
| Boiling point / boiling range          | No data available | None known       |
| Flash point                            | No data available | None known       |
| Evaporation rate                       | No data available | None known       |
| Flammability (solid, gas)              | No data available | None known       |
| Flammability Limit in Air              |                   | None known       |
| Upper flammability or explosive limits | No data available |                  |
| Lower flammability or explosive limits | No data available |                  |
| Vapor pressure                         | No data available | None known       |
| Vapor density                          | No data available | None known       |
| Relative density                       | No data available | None known       |
| Water solubility                       | Miscible in water |                  |
| Solubility in other solvents           | No data available | None known       |
| Partition coefficient                  | No data available | None known       |
| Autoignition temperature               | No data available | None known       |
| Decomposition temperature              | No data available | None known       |
| Kinematic viscosity                    | No data available | None known       |
| Dynamic viscosity                      | No data available | None known       |

### Other information

|                      |                 |
|----------------------|-----------------|
| Oxidizing properties | Not applicable. |
| Explosive properties | Not applicable. |
| Softening point      | Not applicable. |
| Molecular weight     | Not applicable  |
| VOC Content (%)      | Not applicable  |

## 10. Stability and reactivity

|                                    |                                           |
|------------------------------------|-------------------------------------------|
| Reactivity                         | No information available.                 |
| Chemical stability                 | Stable under normal conditions.           |
| Possibility of hazardous reactions | None under normal processing.             |
| Conditions to avoid                | None known based on information supplied. |
| Incompatible materials             | None known based on information supplied. |
| Hazardous decomposition products   | None known based on information supplied. |

## 11. Toxicological information

### Information on likely routes of exposure

Product Information

|              |                                                                                                                                                                                                                             |
|--------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Inhalation   | Specific test data for the substance or mixture is not available.                                                                                                                                                           |
| Eye contact  | Specific test data for the substance or mixture is not available.                                                                                                                                                           |
| Skin contact | May cause sensitization by skin contact. Specific test data for the substance or mixture is not available. Repeated or prolonged skin contact may cause allergic reactions with susceptible persons. (based on components). |
| Ingestion    | Specific test data for the substance or mixture is not available.                                                                                                                                                           |

**Symptoms related to the physical, chemical and toxicological characteristics**

Symptoms Itching. Rashes. Hives.

**Acute toxicity**

Numerical measures of toxicity

**Component Information****Delayed and immediate effects as well as chronic effects from short and long-term exposure**

|                                   |                                                                   |
|-----------------------------------|-------------------------------------------------------------------|
| Interactive effects               | Based on available data, the classification criteria are not met. |
| Skin corrosion/irritation         | Based on available data, the classification criteria are not met. |
| Serious eye damage/eye irritation | Based on available data, the classification criteria are not met. |
| Respiratory or skin sensitization | May cause sensitization by skin contact.                          |
| Germ cell mutagenicity            | Based on available data, the classification criteria are not met. |
| Carcinogenicity                   | Based on available data, the classification criteria are not met. |
| Reproductive toxicity             | Based on available data, the classification criteria are not met. |
| STOT - single exposure            | Based on available data, the classification criteria are not met. |
| STOT - repeated exposure          | Based on available data, the classification criteria are not met. |
| Aspiration hazard                 | Based on available data, the classification criteria are not met. |
| Other information                 | No information available.                                         |

**12. Ecological information**

|                               |                                                    |
|-------------------------------|----------------------------------------------------|
| Ecotoxicity                   | Harmful to aquatic life with long lasting effects. |
| Persistence and degradability | No information available.                          |
| Bioaccumulation               | No information available.                          |
| Other adverse effects         | No information available.                          |

**13. Disposal considerations**

**Waste treatment methods**

|                                            |                                                                                                                 |
|--------------------------------------------|-----------------------------------------------------------------------------------------------------------------|
| <b>Waste from residues/unused products</b> | Dispose of in accordance with local regulations. Dispose of waste in accordance with environmental legislation. |
| <b>Contaminated packaging</b>              | Do not reuse empty containers.                                                                                  |

**14. Transport information**

|                                                                                 |                          |
|---------------------------------------------------------------------------------|--------------------------|
| <b>MEX</b>                                                                      | Not regulated            |
| <b>Transport in bulk according to Annex II of MARPOL 73/78 and the IBC Code</b> | No information available |
| <b>TDG</b>                                                                      | Not regulated            |
| <b>DOT</b>                                                                      | Not regulated            |
| <b>IATA</b>                                                                     | Not regulated            |
| <b>IMDG</b>                                                                     | Not regulated            |

**15. Regulatory information****Safety, health and environmental regulations/legislation specific for the substance or mixture****International Regulations**

**The Montreal Protocol on Substances that Deplete the Ozone Layer** Not applicable

**The Stockholm Convention on Persistent Organic Pollutants** Not applicable

**The Rotterdam Convention** Not applicable

**International Inventories**

Contact supplier for inventory compliance status

**16. Other information**

|             |                         |                       |                           |                                           |
|-------------|-------------------------|-----------------------|---------------------------|-------------------------------------------|
| <b>NFPA</b> | <b>Health hazards</b> 2 | <b>Flammability</b> 0 | <b>Instability</b> 0      | <b>Physical and chemical properties</b> - |
| <b>HMIS</b> | <b>Health hazards</b> 2 | <b>Flammability</b> 0 | <b>Physical hazards</b> 0 | <b>Personal protection</b> X              |

**Key or legend to abbreviations and acronyms used in the safety data sheet****Legend Section 8: EXPOSURE CONTROLS/PERSONAL PROTECTION**

|         |                             |      |                                  |
|---------|-----------------------------|------|----------------------------------|
| TWA     | TWA (time-weighted average) | STEL | STEL (Short Term Exposure Limit) |
| Ceiling | Maximum limit value         | *    | Skin designation                 |

**Key literature references and sources for data used to compile the SDS**

Agency for Toxic Substances and Disease Registry (ATSDR)  
 U.S. Environmental Protection Agency ChemView Database  
 European Food Safety Authority (EFSA)  
 EPA (Environmental Protection Agency)  
 Acute Exposure Guideline Level(s) (AELG(s))  
 U.S. Environmental Protection Agency Federal Insecticide, Fungicide, and Rodenticide Act  
 U.S. Environmental Protection Agency High Production Volume Chemicals  
 Food Research Journal  
 Hazardous Substance Database

International Uniform Chemical Information Database (IUCLID)  
Japan GHS Classification  
Australia National Industrial Chemicals Notification and Assessment Scheme (NICNAS)  
NIOSH (National Institute for Occupational Safety and Health)  
National Library of Medicine's ChemID Plus (NLM CIP)  
National Library of Medicine's PubMed database (NLM PUBMED)  
National Toxicology Program (NTP)  
New Zealand's Chemical Classification and Information Database (CCID)  
Organization for Economic Co-operation and Development Environment, Health, and Safety Publications  
Organization for Economic Co-operation and Development High Production Volume Chemicals Program  
Organization for Economic Co-operation and Development Screening Information Data Set  
RTECS (Registry of Toxic Effects of Chemical Substances)  
World Health Organization

**Prepared By** Bio-Rad Laboratories, Environmental Health and Safety.

**Revision date** 03-Mar-2022

**Revision Note** Significant changes throughout SDS. Review all sections.

**NOM-018-STPS-2015**

**The information is believed to be accurate, but it is not exhaustive and must be used only as guidance. It is based on the current state of knowledge of the chemical substance or mixture and is applicable to the appropriate safety precautions for the product.**

**Disclaimer**

**The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text.**

**End of Safety Data Sheet**

# SAFETY DATA SHEET



Revision date 03-Mar-2022

Revision Number 1

## 1. Identification

### Product identifier

**Product Name** R6 - Conjugate 1 (15 ml)

### Other means of identification

**Catalog Number(s)** 7256K

### Recommended use of the chemical and restrictions on use

**Recommended use** Restricted to professional users In vitro diagnostic

### Details of the supplier of the safety data sheet

#### Corporate Headquarters

Bio-Rad Laboratories Inc.  
1000 Alfred Nobel Drive  
Hercules, CA 94547  
USA

#### Manufacturer Address

Bio-Rad  
3 boulevard Raymond Poincaré  
92430 Marnes-la-Coquette  
France  
e-mail: fds-msds.fr@bio-rad.com

#### Legal Entity / Contact Address

Bio-Rad Laboratories  
Clinical Diagnostics  
4000 Alfred Nobel Drive  
Hercules, California 94547  
USA

**Technical Service** 1-800-224-6723  
TechSupport.USSD@bio-rad.com

### Emergency telephone number

**24 Hour Emergency Phone Number** CHEMTREC USA: 1 (800) 424-9300

## 2. Hazard(s) identification

### Classification

This chemical is not considered hazardous by the 2012 OSHA Hazard Communication Standard (29 CFR 1910.1200)

### Hazards not otherwise classified (HNOC)

Not applicable

### Label elements

#### Hazard statements

This chemical is not considered hazardous by the 2012 OSHA Hazard Communication Standard (29 CFR 1910.1200)

The product contains no substances which at their given concentration, are considered to be hazardous to health.

**Appearance** Liquid

**Physical state** Liquid

**Odor** Odorless

### **Precautionary Statements - Disposal**

Dispose of contents/container in accordance with local, regional, national, and international regulations as applicable



**Other information**

Contains animal source material. Harmful to aquatic life with long lasting effects. Harmful to aquatic life.

**3. Composition/information on ingredients****Substance**

Not applicable.

**Mixture**

| Chemical name      | CAS No     | Weight-%    | Trade secret |
|--------------------|------------|-------------|--------------|
| 1,2,3-Propanetriol | 56-81-5    | 10 - 20     | *            |
| Sodium chloride    | 7647-14-5  | 2.5 - 5     | *            |
| Sodium azide       | 26628-22-8 | 0.1 - 0.299 | *            |

\*The exact percentage (concentration) of composition has been withheld as a trade secret.

**4. First-aid measures****Description of first aid measures**

|              |                                                                                                                      |
|--------------|----------------------------------------------------------------------------------------------------------------------|
| Inhalation   | Remove to fresh air.                                                                                                 |
| Eye contact  | Rinse thoroughly with plenty of water for at least 15 minutes, lifting lower and upper eyelids. Consult a physician. |
| Skin contact | Wash skin with soap and water.                                                                                       |
| Ingestion    | Rinse mouth.                                                                                                         |

**Most important symptoms and effects, both acute and delayed**

|          |                           |
|----------|---------------------------|
| Symptoms | No information available. |
|----------|---------------------------|

**Indication of any immediate medical attention and special treatment needed**

|                    |                        |
|--------------------|------------------------|
| Note to physicians | Treat symptomatically. |
|--------------------|------------------------|

**5. Fire-fighting measures**

|                              |                                                                                                         |
|------------------------------|---------------------------------------------------------------------------------------------------------|
| Suitable Extinguishing Media | Use extinguishing measures that are appropriate to local circumstances and the surrounding environment. |
|------------------------------|---------------------------------------------------------------------------------------------------------|

|                                            |                           |
|--------------------------------------------|---------------------------|
| Specific hazards arising from the chemical | No information available. |
|--------------------------------------------|---------------------------|

|                                  |       |
|----------------------------------|-------|
| Explosion data                   |       |
| Sensitivity to mechanical impact | None. |

|                                 |       |
|---------------------------------|-------|
| Sensitivity to static discharge | None. |
|---------------------------------|-------|

|                                                                |                                                                                                                                    |
|----------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------|
| Special protective equipment and precautions for fire-fighters | Firefighters should wear self-contained breathing apparatus and full firefighting turnout gear. Use personal protection equipment. |
|----------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------|

**6. Accidental release measures**

**Personal precautions, protective equipment and emergency procedures**

**Personal precautions** Ensure adequate ventilation.

**Methods and material for containment and cleaning up**

**Methods for containment** Prevent further leakage or spillage if safe to do so.

**Methods for cleaning up** Pick up and transfer to properly labeled containers.

## 7. Handling and storage

**Precautions for safe handling**

**Advice on safe handling** Handle in accordance with good industrial hygiene and safety practice.

**Conditions for safe storage, including any incompatibilities**

**Storage Conditions** Store according to product and label instructions.

## 8. Exposure controls/personal protection

**Control parameters**

**Exposure Limits** The following ingredients are the only ingredients of the product above the cut-off level (or level that contributes to the hazard classification of the mixture) which have an exposure limit applicable in the region for which this safety data sheet is intended or other recommended limit. At this time, the other relevant constituents have no known exposure limits from the sources listed here.

| Chemical name                 | ACGIH TLV                                                                              | OSHA PEL                                                                                                                                                                                                                               | NIOSH                                                                               |
|-------------------------------|----------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
| 1,2,3-Propanetriol<br>56-81-5 | -                                                                                      | TWA: 15 mg/m <sup>3</sup> mist, total particulate<br>TWA: 5 mg/m <sup>3</sup> mist, respirable fraction<br>(vacated) TWA: 10 mg/m <sup>3</sup> mist, total particulate<br>(vacated) TWA: 5 mg/m <sup>3</sup> mist, respirable fraction | -                                                                                   |
| Sodium azide<br>26628-22-8    | Ceiling: 0.29 mg/m <sup>3</sup> Sodium azide<br>Ceiling: 0.11 ppm Hydrazoic acid vapor | (vacated) S*<br>(vacated) Ceiling: 0.1 ppm HN <sub>3</sub><br>(vacated) Ceiling: 0.3 mg/m <sup>3</sup> NaN <sub>3</sub>                                                                                                                | Ceiling: 0.1 ppm HN <sub>3</sub><br>Ceiling: 0.3 mg/m <sup>3</sup> NaN <sub>3</sub> |

**Appropriate engineering controls**

**Engineering controls** Showers  
Eyewash stations  
Ventilation systems.

**Individual protection measures, such as personal protective equipment**

**Eye/face protection** No special protective equipment required.

**Skin and body protection** No special protective equipment required.

**Respiratory protection** No protective equipment is needed under normal use conditions. If exposure limits are exceeded or irritation is experienced, ventilation and evacuation may be required.

**General hygiene considerations** Handle in accordance with good industrial hygiene and safety practice.

## 9. Physical and chemical properties

### Information on basic physical and chemical properties

|                |                          |
|----------------|--------------------------|
| Physical state | Liquid                   |
| Appearance     | Liquid                   |
| Color          | violet                   |
| Odor           | Odorless                 |
| Odor threshold | No information available |

| <u>Property</u>                        | <u>Values</u>      | <u>Remarks • Method</u> |
|----------------------------------------|--------------------|-------------------------|
| pH                                     |                    | None known              |
| Melting point / freezing point         | No data available  | None known              |
| Boiling point / boiling range          | No data available  | None known              |
| Flash point                            | No data available  | None known              |
| Evaporation rate                       | No data available  | None known              |
| Flammability (solid, gas)              | No data available  | None known              |
| Flammability Limit in Air              |                    | None known              |
| Upper flammability or explosive limits | No data available  |                         |
| Lower flammability or explosive limits | No data available  |                         |
| Vapor pressure                         | No data available  | None known              |
| Vapor density                          | No data available  | None known              |
| Relative density                       | No data available  | None known              |
| Water solubility                       | Miscible in water  |                         |
| Solubility(ies)                        | No data available  | None known              |
| Partition coefficient                  | No data available  | None known              |
| Autoignition temperature               | 392.78 °C / 739 °F |                         |
| Decomposition temperature              |                    | None known              |
| Kinematic viscosity                    | No data available  | None known              |
| Dynamic viscosity                      | No data available  | None known              |

### Other information

|                      |                          |
|----------------------|--------------------------|
| Explosive properties | No information available |
| Oxidizing properties | No information available |
| Softening point      | No information available |
| Molecular weight     | No information available |
| VOC Content (%)      | No information available |
| Liquid Density       | No information available |
| Bulk density         | No information available |

## 10. Stability and reactivity

**Reactivity** No information available.

**Chemical stability** Stable under normal conditions.

**Possibility of hazardous reactions** Avoid contact with metals. This product contains sodium azide. Sodium azide can react with copper, brass, lead, and solder in piping systems to form explosive compounds and toxic gases.

**Conditions to avoid** None known based on information supplied.

**Incompatible materials** Metals.

**Hazardous decomposition products** None known based on information supplied.

## 11. Toxicological information

### Information on likely routes of exposure

|                     |                                                                   |
|---------------------|-------------------------------------------------------------------|
| <b>Inhalation</b>   | Specific test data for the substance or mixture is not available. |
| <b>Eye contact</b>  | Specific test data for the substance or mixture is not available. |
| <b>Skin contact</b> | Specific test data for the substance or mixture is not available. |
| <b>Ingestion</b>    | Specific test data for the substance or mixture is not available. |

### Symptoms related to the physical, chemical and toxicological characteristics

|                 |                           |
|-----------------|---------------------------|
| <b>Symptoms</b> | No information available. |
|-----------------|---------------------------|

### Acute toxicity

### **Numerical measures of toxicity**

The following values are calculated based on chapter 3.1 of the GHS document

|                        |                 |
|------------------------|-----------------|
| <b>ATEmix (oral)</b>   | 49,984.80 mg/kg |
| <b>ATEmix (dermal)</b> | 82,631.70 mg/kg |

### **Component Information**

| Chemical name                 | Oral LD50             | Dermal LD50              | Inhalation LC50               |
|-------------------------------|-----------------------|--------------------------|-------------------------------|
| 1,2,3-Propanetriol<br>56-81-5 | = 12600 mg/kg ( Rat ) | > 10 g/kg ( Rabbit )     | > 2.75 mg/L ( Rat ) 4 h       |
| Sodium chloride<br>7647-14-5  | = 3 g/kg ( Rat )      | > 10000 mg/kg ( Rabbit ) | > 42 mg/L ( Rat ) 1 h         |
| Sodium azide<br>26628-22-8    | = 27 mg/kg ( Rat )    | = 20 mg/kg ( Rabbit )    | 0.054 - 0.52 mg/L ( Rat ) 4 h |

### Delayed and immediate effects as well as chronic effects from short and long-term exposure

|                                          |                           |
|------------------------------------------|---------------------------|
| <b>Skin corrosion/irritation</b>         | No information available. |
| <b>Serious eye damage/eye irritation</b> | No information available. |
| <b>Respiratory or skin sensitization</b> | No information available. |
| <b>Germ cell mutagenicity</b>            | No information available. |
| <b>Carcinogenicity</b>                   | No information available. |
| <b>Reproductive toxicity</b>             | No information available. |
| <b>STOT - single exposure</b>            | No information available. |
| <b>STOT - repeated exposure</b>          | No information available. |

**Target organ effects** Kidney, Respiratory system, Eyes, Skin.

**Aspiration hazard** No information available.

**Other adverse effects** No information available.

**Interactive effects** No information available.

## 12. Ecological information

**Ecotoxicity** Harmful to aquatic life with long lasting effects.

| Chemical name                 | Algae/aquatic plants | Fish                                                                                                                                                                                                                                                                                                      | Toxicity to microorganisms | Crustacea                                                                            |
|-------------------------------|----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------|--------------------------------------------------------------------------------------|
| 1,2,3-Propanetriol<br>56-81-5 | -                    | LC50: 51 - 57mL/L (96h, Oncorhynchus mykiss)                                                                                                                                                                                                                                                              | -                          | -                                                                                    |
| Sodium chloride<br>7647-14-5  | -                    | LC50: 4747 - 7824mg/L (96h, Oncorhynchus mykiss)<br>LC50: 5560 - 6080mg/L (96h, Lepomis macrochirus)<br>LC50: 6020 - 7070mg/L (96h, Pimephales promelas)<br>LC50: 6420 - 6700mg/L (96h, Pimephales promelas)<br>LC50: =12946mg/L (96h, Lepomis macrochirus)<br>LC50: =7050mg/L (96h, Pimephales promelas) | -                          | EC50: 340.7 - 469.2mg/L (48h, Daphnia magna)<br>EC50: =1000mg/L (48h, Daphnia magna) |
| Sodium azide<br>26628-22-8    | -                    | LC50: =0.7mg/L (96h, Lepomis macrochirus)<br>LC50: =0.8mg/L (96h, Oncorhynchus mykiss)<br>LC50: =5.46mg/L (96h, Pimephales promelas)                                                                                                                                                                      | -                          | -                                                                                    |

**Persistence and degradability** No information available.

### Bioaccumulation

#### Component Information

| Chemical name                 | Partition coefficient |
|-------------------------------|-----------------------|
| 1,2,3-Propanetriol<br>56-81-5 | -1.76                 |

**Other adverse effects** No information available.

## 13. Disposal considerations

### Waste treatment methods

#### Waste from residues/unused products

Dispose of in accordance with local regulations. Dispose of waste in accordance with environmental legislation. Flush pipes with water frequently if discarding solutions containing sodium azide into metal piping systems.

**Contaminated packaging** Do not reuse empty containers.

**California Hazardous Waste Status** This product contains one or more substances that are listed with the State of California as a hazardous waste.

## 14. Transport information

**DOT** Not regulated

**TDG** Not regulated

**MEX** Not regulated

**IATA** Not regulated

**IMDG** Not regulated

## 15. Regulatory information

**International Inventories** Contact supplier for inventory compliance status

### US Federal Regulations

#### SARA 313

Section 313 of Title III of the Superfund Amendments and Reauthorization Act of 1986 (SARA). This product contains a chemical or chemicals which are subject to the reporting requirements of the Act and Title 40 of the Code of Federal Regulations, Part 372.

| Chemical name             | SARA 313 - Threshold Values % |
|---------------------------|-------------------------------|
| Sodium azide - 26628-22-8 | 1.0                           |

#### SARA 311/312 Hazard Categories

Should this product meet EPCRA 311/312 Tier reporting criteria at 40 CFR 370, refer to Section 2 of this SDS for appropriate classifications.

#### CWA (Clean Water Act)

This product does not contain any substances regulated as pollutants pursuant to the Clean Water Act (40 CFR 122.21 and 40 CFR 122.42).

#### CERCLA

This material, as supplied, contains one or more substances regulated as a hazardous substance under the Comprehensive Environmental Response Compensation and Liability Act (CERCLA) (40 CFR 302).

| Chemical name              | Hazardous Substances RQs | Extremely Hazardous Substances RQs | Reportable Quantity (RQ)                  |
|----------------------------|--------------------------|------------------------------------|-------------------------------------------|
| Sodium azide<br>26628-22-8 | 1000 lb                  | 1000 lb                            | RQ 1000 lb final RQ<br>RQ 454 kg final RQ |

### US State Regulations

#### California Proposition 65

This product does not contain any Proposition 65 chemicals.

### U.S. State Right-to-Know Regulations

| Chemical name      | New Jersey | Massachusetts | Pennsylvania |
|--------------------|------------|---------------|--------------|
| Water<br>7732-18-5 | -          | -             | X            |

**R6 - Conjugate 1 (15 ml)**

Revision date 03-Mar-2022

|                                        |   |   |   |
|----------------------------------------|---|---|---|
| 1,2,3-Propanetriol<br>56-81-5          | X | X | X |
| Sodium azide<br>26628-22-8             | X | X | X |
| Propane, 1-bromo-3-chloro-<br>109-70-6 | X | - | - |

**U.S. EPA Label Information****EPA Pesticide Registration Number** Not applicable**16. Other information**

|             |                         |                       |                           |                              |
|-------------|-------------------------|-----------------------|---------------------------|------------------------------|
| <b>NFPA</b> | <b>Health hazards</b> 0 | <b>Flammability</b> 1 | <b>Instability</b> 0      | <b>Special hazards</b> -     |
| <b>HMIS</b> | <b>Health hazards</b> 0 | <b>Flammability</b> 1 | <b>Physical hazards</b> 0 | <b>Personal protection</b> X |

**Key or legend to abbreviations and acronyms used in the safety data sheet****Legend Section 8: EXPOSURE CONTROLS/PERSONAL PROTECTION**

|         |                             |      |                                  |
|---------|-----------------------------|------|----------------------------------|
| TWA     | TWA (time-weighted average) | STEL | STEL (Short Term Exposure Limit) |
| Ceiling | Maximum limit value         | *    | Skin designation                 |

**Key literature references and sources for data used to compile the SDS**

Agency for Toxic Substances and Disease Registry (ATSDR)  
 U.S. Environmental Protection Agency ChemView Database  
 European Food Safety Authority (EFSA)  
 EPA (Environmental Protection Agency)  
 Acute Exposure Guideline Level(s) (AEGL(s))  
 U.S. Environmental Protection Agency Federal Insecticide, Fungicide, and Rodenticide Act  
 U.S. Environmental Protection Agency High Production Volume Chemicals  
 Food Research Journal  
 Hazardous Substance Database  
 International Uniform Chemical Information Database (IUCLID)  
 National Institute of Technology and Evaluation (NITE)  
 Australia National Industrial Chemicals Notification and Assessment Scheme (NICNAS)  
 NIOSH (National Institute for Occupational Safety and Health)  
 National Library of Medicine's ChemID Plus (NLM CIP)  
 National Library of Medicine's PubMed database (NLM PUBMED)  
 National Toxicology Program (NTP)  
 New Zealand's Chemical Classification and Information Database (CCID)  
 Organization for Economic Co-operation and Development Environment, Health, and Safety Publications  
 Organization for Economic Co-operation and Development High Production Volume Chemicals Program  
 Organization for Economic Co-operation and Development Screening Information Data Set  
 World Health Organization

**Revision date** 03-Mar-2022**Revision Note** Significant changes throughout SDS. Review all sections.**Disclaimer**

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text.

**End of Safety Data Sheet**

# SAFETY DATA SHEET



Revision date 03-Mar-2022

Revision Number 1

## 1. Identification

### Product identifier

**Product Name** R7 - Conjugate 2 (15 ml)

### Other means of identification

**Catalog Number(s)** 7256N

### Recommended use of the chemical and restrictions on use

**Recommended use** Restricted to professional users In vitro diagnostic

### Details of the supplier of the safety data sheet

#### Corporate Headquarters

Bio-Rad Laboratories Inc.  
1000 Alfred Nobel Drive  
Hercules, CA 94547  
USA

#### Manufacturer Address

Bio-Rad  
3 boulevard Raymond Poincaré  
92430 Marnes-la-Coquette  
France  
e-mail: fds-msds.fr@bio-rad.com

#### Legal Entity / Contact Address

Bio-Rad Laboratories  
Clinical Diagnostics  
4000 Alfred Nobel Drive  
Hercules, California 94547  
USA

**Technical Service** 1-800-224-6723  
TechSupport.USSD@bio-rad.com

### Emergency telephone number

**24 Hour Emergency Phone Number** CHEMTREC USA: 1 (800) 424-9300

## 2. Hazard(s) identification

### Classification

|                    |             |
|--------------------|-------------|
| Skin sensitization | Category 1A |
|--------------------|-------------|

### Hazards not otherwise classified (HNOC)

Not applicable

### Label elements

#### Warning

#### Hazard statements

May cause an allergic skin reaction





|                          |                              |                      |
|--------------------------|------------------------------|----------------------|
| <b>Appearance</b> Liquid | <b>Physical state</b> Liquid | <b>Odor</b> Odorless |
|--------------------------|------------------------------|----------------------|

**Precautionary Statements - Prevention**

Avoid breathing dust/fume/gas/mist/vapors/spray  
 Contaminated work clothing must not be allowed out of the workplace  
 Wear protective gloves

IF ON SKIN: Wash with plenty of water and soap  
 If skin irritation or rash occurs: Get medical advice/attention  
 Wash contaminated clothing before reuse

**Precautionary Statements - Disposal**

Dispose of contents/container in accordance with local, regional, national, and international regulations as applicable

**Other information**

Harmful to aquatic life with long lasting effects. Harmful to aquatic life. Contains animal source material.

### 3. Composition/information on ingredients

**Substance**

Not applicable.

**Mixture**

| Chemical name                                                                    | CAS No     | Weight-%     | Trade secret |
|----------------------------------------------------------------------------------|------------|--------------|--------------|
| 1,2,3-Propanetriol                                                               | 56-81-5    | 20 - 35      | *            |
| Sodium chloride                                                                  | 7647-14-5  | 10 - 20      | *            |
| 5-Chloro-2-methyl-3(2H)-isothiazolone, mixture with 2-methyl-3(2H)-isothiazolone | 55965-84-9 | 0.01 - 0.099 | *            |

\*The exact percentage (concentration) of composition has been withheld as a trade secret.

### 4. First-aid measures

**Description of first aid measures**

|                       |                                                                                                                                      |
|-----------------------|--------------------------------------------------------------------------------------------------------------------------------------|
| <b>General advice</b> | Show this safety data sheet to the doctor in attendance.                                                                             |
| <b>Inhalation</b>     | Remove to fresh air.                                                                                                                 |
| <b>Eye contact</b>    | Rinse thoroughly with plenty of water for at least 15 minutes, lifting lower and upper eyelids. Consult a physician.                 |
| <b>Skin contact</b>   | Wash with soap and water. May cause an allergic skin reaction. In the case of skin irritation or allergic reactions see a physician. |
| <b>Ingestion</b>      | Rinse mouth.                                                                                                                         |

**Most important symptoms and effects, both acute and delayed**

**Symptoms** Itching. Rashes. Hives.

**Indication of any immediate medical attention and special treatment needed**

**Note to physicians** May cause sensitization in susceptible persons. Treat symptomatically.

### 5. Fire-fighting measures

|                                                                       |                                                                                                                                    |
|-----------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------|
| <b>Suitable Extinguishing Media</b>                                   | Use extinguishing measures that are appropriate to local circumstances and the surrounding environment.                            |
| <b>Specific hazards arising from the chemical</b>                     | Product is or contains a sensitizer. May cause sensitization by skin contact.                                                      |
| <b>Explosion data</b>                                                 |                                                                                                                                    |
| <b>Sensitivity to mechanical impact</b>                               | None.                                                                                                                              |
| <b>Sensitivity to static discharge</b>                                | None.                                                                                                                              |
| <b>Special protective equipment and precautions for fire-fighters</b> | Firefighters should wear self-contained breathing apparatus and full firefighting turnout gear. Use personal protection equipment. |

## 6. Accidental release measures

### Personal precautions, protective equipment and emergency procedures

|                             |                                                                                                                                                                                                          |
|-----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Personal precautions</b> | Avoid contact with skin, eyes or clothing. Ensure adequate ventilation. Use personal protective equipment as required. Evacuate personnel to safe areas. Keep people away from and upwind of spill/leak. |
|-----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

### Methods and material for containment and cleaning up

|                                |                                                       |
|--------------------------------|-------------------------------------------------------|
| <b>Methods for containment</b> | Prevent further leakage or spillage if safe to do so. |
| <b>Methods for cleaning up</b> | Pick up and transfer to properly labeled containers.  |

## 7. Handling and storage

### Precautions for safe handling

|                                |                                                                                                                                                                                                                                                                                                                                    |
|--------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Advice on safe handling</b> | Handle in accordance with good industrial hygiene and safety practice. Avoid contact with skin, eyes or clothing. Ensure adequate ventilation. In case of insufficient ventilation, wear suitable respiratory equipment. Do not eat, drink or smoke when using this product. Take off contaminated clothing and wash before reuse. |
|--------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

### Conditions for safe storage, including any incompatibilities

|                           |                                                                                                                             |
|---------------------------|-----------------------------------------------------------------------------------------------------------------------------|
| <b>Storage Conditions</b> | Keep containers tightly closed in a dry, cool and well-ventilated place. Store according to product and label instructions. |
|---------------------------|-----------------------------------------------------------------------------------------------------------------------------|

## 8. Exposure controls/personal protection

### Control parameters

|                        |                                                                                                                                                                                                                                                                                                                                                                                                            |
|------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Exposure Limits</b> | The following ingredients are the only ingredients of the product above the cut-off level (or level that contributes to the hazard classification of the mixture) which have an exposure limit applicable in the region for which this safety data sheet is intended or other recommended limit. At this time, the other relevant constituents have no known exposure limits from the sources listed here. |
|------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

| Chemical name      | ACGIH TLV | OSHA PEL                              | NIOSH |
|--------------------|-----------|---------------------------------------|-------|
| 1,2,3-Propanetriol | -         | TWA: 15 mg/m <sup>3</sup> mist, total | -     |

|         |  |                                                                                                                                                                                                  |  |
|---------|--|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
| 56-81-5 |  | particulate<br>TWA: 5 mg/m <sup>3</sup> mist, respirable fraction<br>(vacated) TWA: 10 mg/m <sup>3</sup> mist, total particulate<br>(vacated) TWA: 5 mg/m <sup>3</sup> mist, respirable fraction |  |
|---------|--|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|

**Appropriate engineering controls**

**Engineering controls**                      Showers  
                                                          Eyewash stations  
                                                          Ventilation systems.

**Individual protection measures, such as personal protective equipment**

**Eye/face protection**                      Wear safety glasses with side shields (or goggles).

**Hand protection**                              Wear suitable gloves.

**Skin and body protection**                      Wear suitable protective clothing.

**Respiratory protection**                      No protective equipment is needed under normal use conditions. If exposure limits are exceeded or irritation is experienced, ventilation and evacuation may be required.

**General hygiene considerations**                      Handle in accordance with good industrial hygiene and safety practice.

## 9. Physical and chemical properties

**Information on basic physical and chemical properties**

**Physical state**                              Liquid  
**Appearance**                                Liquid  
**Color**                                          green  
**Odor**                                          Odorless  
**Odor threshold**                              No information available

| <b><u>Property</u></b>                        | <b><u>Values</u></b> | <b><u>Remarks • Method</u></b> |
|-----------------------------------------------|----------------------|--------------------------------|
| <b>pH</b>                                     |                      | None known                     |
| <b>Melting point / freezing point</b>         | No data available    | None known                     |
| <b>Boiling point / boiling range</b>          | No data available    | None known                     |
| <b>Flash point</b>                            | No data available    | None known                     |
| <b>Evaporation rate</b>                       | No data available    | None known                     |
| <b>Flammability (solid, gas)</b>              | No data available    | None known                     |
| <b>Flammability Limit in Air</b>              |                      | None known                     |
| <b>Upper flammability or explosive limits</b> | No data available    |                                |
| <b>Lower flammability or explosive limits</b> | No data available    |                                |
| <b>Vapor pressure</b>                         | No data available    | None known                     |
| <b>Vapor density</b>                          | No data available    | None known                     |
| <b>Relative density</b>                       | No data available    | None known                     |
| <b>Water solubility</b>                       | Miscible in water    |                                |
| <b>Solubility(ies)</b>                        | No data available    | None known                     |
| <b>Partition coefficient</b>                  | No data available    | None known                     |
| <b>Autoignition temperature</b>               | 392.78               | None known                     |
| <b>Decomposition temperature</b>              |                      | None known                     |
| <b>Kinematic viscosity</b>                    | No data available    | None known                     |
| <b>Dynamic viscosity</b>                      | No data available    | None known                     |

**Other information**

**Explosive properties**                      No information available

|                             |                          |
|-----------------------------|--------------------------|
| <b>Oxidizing properties</b> | No information available |
| <b>Softening point</b>      | No information available |
| <b>Molecular weight</b>     | No information available |
| <b>VOC Content (%)</b>      | No information available |
| <b>Liquid Density</b>       | No information available |
| <b>Bulk density</b>         | No information available |

## 10. Stability and reactivity

|                                           |                                           |
|-------------------------------------------|-------------------------------------------|
| <b>Reactivity</b>                         | No information available.                 |
| <b>Chemical stability</b>                 | Stable under normal conditions.           |
| <b>Possibility of hazardous reactions</b> | None under normal processing.             |
| <b>Conditions to avoid</b>                | None known based on information supplied. |
| <b>Incompatible materials</b>             | None known based on information supplied. |
| <b>Hazardous decomposition products</b>   | None known based on information supplied. |

## 11. Toxicological information

### Information on likely routes of exposure

#### Product Information

|                     |                                                                                                                                                                                                                             |
|---------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Inhalation</b>   | Specific test data for the substance or mixture is not available.                                                                                                                                                           |
| <b>Eye contact</b>  | Specific test data for the substance or mixture is not available.                                                                                                                                                           |
| <b>Skin contact</b> | May cause sensitization by skin contact. Specific test data for the substance or mixture is not available. Repeated or prolonged skin contact may cause allergic reactions with susceptible persons. (based on components). |
| <b>Ingestion</b>    | Specific test data for the substance or mixture is not available.                                                                                                                                                           |

### Symptoms related to the physical, chemical and toxicological characteristics

|                 |                         |
|-----------------|-------------------------|
| <b>Symptoms</b> | Itching. Rashes. Hives. |
|-----------------|-------------------------|

### Acute toxicity

#### Numerical measures of toxicity

The following values are calculated based on chapter 3.1 of the GHS document

ATEmix (oral) 25,000.00 mg/kg

#### Component Information

| Chemical name                                                                                     | Oral LD50             | Dermal LD50              | Inhalation LC50         |
|---------------------------------------------------------------------------------------------------|-----------------------|--------------------------|-------------------------|
| 1,2,3-Propanetriol<br>56-81-5                                                                     | = 12600 mg/kg ( Rat ) | > 10 g/kg ( Rabbit )     | > 2.75 mg/L ( Rat ) 4 h |
| Sodium chloride<br>7647-14-5                                                                      | = 3 g/kg ( Rat )      | > 10000 mg/kg ( Rabbit ) | > 42 mg/L ( Rat ) 1 h   |
| 5-Chloro-2-methyl-3(2H)-isothiazolone, mixture with<br>2-methyl-3(2H)-isothiazolone<br>55965-84-9 | = 53 mg/kg ( Rat )    | = 87.12 mg/kg ( Rabbit ) | -                       |

### Delayed and immediate effects as well as chronic effects from short and long-term exposure

|                                          |                                          |
|------------------------------------------|------------------------------------------|
| <b>Skin corrosion/irritation</b>         | No information available.                |
| <b>Serious eye damage/eye irritation</b> | No information available.                |
| <b>Respiratory or skin sensitization</b> | May cause sensitization by skin contact. |
| <b>Germ cell mutagenicity</b>            | No information available.                |
| <b>Carcinogenicity</b>                   | No information available.                |
| <b>Reproductive toxicity</b>             | No information available.                |
| <b>STOT - single exposure</b>            | No information available.                |
| <b>STOT - repeated exposure</b>          | No information available.                |
| <b>Target organ effects</b>              | Kidney, Respiratory system, Eyes, Skin.  |
| <b>Aspiration hazard</b>                 | No information available.                |
| <b>Other adverse effects</b>             | No information available.                |
| <b>Interactive effects</b>               | No information available.                |

## 12. Ecological information

**Ecotoxicity** Harmful to aquatic life with long lasting effects.

| Chemical name                 | Algae/aquatic plants | Fish                                                                                                                                                                                                                                                                                                      | Toxicity to microorganisms | Crustacea                                                                            |
|-------------------------------|----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------|--------------------------------------------------------------------------------------|
| 1,2,3-Propanetriol<br>56-81-5 | -                    | LC50: 51 - 57mL/L (96h, Oncorhynchus mykiss)                                                                                                                                                                                                                                                              | -                          | -                                                                                    |
| Sodium chloride<br>7647-14-5  | -                    | LC50: 4747 - 7824mg/L (96h, Oncorhynchus mykiss)<br>LC50: 5560 - 6080mg/L (96h, Lepomis macrochirus)<br>LC50: 6020 - 7070mg/L (96h, Pimephales promelas)<br>LC50: 6420 - 6700mg/L (96h, Pimephales promelas)<br>LC50: =12946mg/L (96h, Lepomis macrochirus)<br>LC50: =7050mg/L (96h, Pimephales promelas) | -                          | EC50: 340.7 - 469.2mg/L (48h, Daphnia magna)<br>EC50: =1000mg/L (48h, Daphnia magna) |

**Persistence and degradability** No information available.

**Bioaccumulation****Component Information**

| Chemical name                 | Partition coefficient |
|-------------------------------|-----------------------|
| 1,2,3-Propanetriol<br>56-81-5 | -1.76                 |

**Other adverse effects** No information available.

**13. Disposal considerations****Waste treatment methods**

**Waste from residues/unused products** Dispose of in accordance with local regulations. Dispose of waste in accordance with environmental legislation.

**Contaminated packaging** Do not reuse empty containers.

**14. Transport information**

**DOT** Not regulated

**TDG** Not regulated

**MEX** Not regulated

**IATA** Not regulated

**IMDG** Not regulated

**15. Regulatory information**

**International Inventories** Contact supplier for inventory compliance status

**US Federal Regulations****SARA 313**

Section 313 of Title III of the Superfund Amendments and Reauthorization Act of 1986 (SARA). This product does not contain any chemicals which are subject to the reporting requirements of the Act and Title 40 of the Code of Federal Regulations, Part 372.

**SARA 311/312 Hazard Categories**

Should this product meet EPCRA 311/312 Tier reporting criteria at 40 CFR 370, refer to Section 2 of this SDS for appropriate classifications.

**CWA (Clean Water Act)**

This product does not contain any substances regulated as pollutants pursuant to the Clean Water Act (40 CFR 122.21 and 40 CFR 122.42).

**CERCLA**

This material, as supplied, does not contain any substances regulated as hazardous substances under the Comprehensive Environmental Response Compensation and Liability Act (CERCLA) (40 CFR 302) or the Superfund Amendments and Reauthorization Act (SARA) (40 CFR 355). There may be specific reporting requirements at the local, regional, or state level pertaining to releases of this material.

**US State Regulations**

**California Proposition 65**

This product does not contain any Proposition 65 chemicals.

**U.S. State Right-to-Know Regulations**

| Chemical name                 | New Jersey | Massachusetts | Pennsylvania |
|-------------------------------|------------|---------------|--------------|
| Water<br>7732-18-5            | -          | -             | X            |
| 1,2,3-Propanetriol<br>56-81-5 | X          | X             | X            |

**U.S. EPA Label Information**

**EPA Pesticide Registration Number** Not applicable

|                              |
|------------------------------|
| <b>16. Other information</b> |
|------------------------------|

|             |                         |                       |                           |                              |
|-------------|-------------------------|-----------------------|---------------------------|------------------------------|
| <b>NFPA</b> | <b>Health hazards</b> 2 | <b>Flammability</b> 1 | <b>Instability</b> 0      | <b>Special hazards</b> -     |
| <b>HMIS</b> | <b>Health hazards</b> 2 | <b>Flammability</b> 1 | <b>Physical hazards</b> 0 | <b>Personal protection</b> X |

**Key or legend to abbreviations and acronyms used in the safety data sheet****Legend Section 8: EXPOSURE CONTROLS/PERSONAL PROTECTION**

|         |                             |      |                                  |
|---------|-----------------------------|------|----------------------------------|
| TWA     | TWA (time-weighted average) | STEL | STEL (Short Term Exposure Limit) |
| Ceiling | Maximum limit value         | *    | Skin designation                 |

**Key literature references and sources for data used to compile the SDS**

Agency for Toxic Substances and Disease Registry (ATSDR)  
 U.S. Environmental Protection Agency ChemView Database  
 European Food Safety Authority (EFSA)  
 EPA (Environmental Protection Agency)  
 Acute Exposure Guideline Level(s) (AEGL(s))  
 U.S. Environmental Protection Agency Federal Insecticide, Fungicide, and Rodenticide Act  
 U.S. Environmental Protection Agency High Production Volume Chemicals  
 Food Research Journal  
 Hazardous Substance Database  
 International Uniform Chemical Information Database (IUCLID)  
 National Institute of Technology and Evaluation (NITE)  
 Australia National Industrial Chemicals Notification and Assessment Scheme (NICNAS)  
 NIOSH (National Institute for Occupational Safety and Health)  
 National Library of Medicine's ChemID Plus (NLM CIP)  
 National Library of Medicine's PubMed database (NLM PUBMED)  
 National Toxicology Program (NTP)  
 New Zealand's Chemical Classification and Information Database (CCID)  
 Organization for Economic Co-operation and Development Environment, Health, and Safety Publications  
 Organization for Economic Co-operation and Development High Production Volume Chemicals Program  
 Organization for Economic Co-operation and Development Screening Information Data Set  
 World Health Organization

**Revision date** 03-Mar-2022

**Revision Note** Significant changes throughout SDS. Review all sections.

**Disclaimer**

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text.

**End of Safety Data Sheet**